# Session 01: Loan EMI Calculator, Claude Code development session

This PDF is the conversation. Each run of tool work is shown as one line; the full tool calls and outputs are in `session-01-claude-code.md` and `raw/`, and the subagent runs in `subagents/`.

One Claude Code session (VS Code extension, Opus 5.5 as the developer) covering the planning phase through the first deploy. Owner messages are verbatim. Tool calls are collapsed and shortened; subagent runs link to their own transcripts. See `README.md` in this folder for what was left out.

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### 👱 Muhammad Arslan (owner) · 2026-10-17 20:44 PKT

> Can you go through the whole folder and share your thoughts?

### 🤖 Claude · 2026-10-17 20:44 PKT

I'll start by mapping the folder, then read the key documents.

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-17 20:44 PKT

It's docs-only so far (~1,150 lines, no code, no git repo yet). I'll read all of it.

🔧 5 tool call(s): Bash ×4, Read

### 🤖 Claude · 2026-10-17 20:46 PKT

Python 3.13 and Node 24.16 are available. Git is installed, but this folder is not a git repo yet, which matters later. Next, the verification script:

🔧 2 tool call(s): Write, Bash

### 🤖 Claude · 2026-10-17 20:47 PKT

## Overall

This is a strong pack. The roles lead to jobs, the jobs lead to stories, and every AC has a fixed expected value. The decision log shows its reasoning. The process splits maker and checker and freezes the tests at G3. The hard part, the rounding method, was actually researched instead of assumed. It's ready to build on once the issues below are settled at G1.

There is no code yet. The folder holds 11 files of docs and agent definitions, and it isn't a git repo.

## What I checked myself (independent Python `decimal`/`fractions`, scratchpad only)

- **All 7 reference loans match.** So do the L-A rows 1, 6 and 12, the 88,848.78 months (9, 10, 12) and the L-Z3 rows.
- **Both §5.3 failure claims are true.** Rs 10,000 at 36% over 359 months pays off at month 352, and Rs 10,000.82 at 30% gives a final instalment of 1,767.73.
- **Invariants 1–6 held on 400 random valid loans.** The largest gap from the headline payment was exactly 0.01.

- **Precision is safe.** I suspected that rounding `r = rate/1200` to 40 digits could flip half-cent interest values. It doesn't: the 40-digit rounding of the product absorbs the error. I found 0 differences from exact rational arithmetic.
- **All text pairs in the §8.4 colour tokens pass 4.5:1.** The lowest is light error-on-surface at 6.13.

## Issues, most important first

### Fix before Phase 1

1. **Most of the CLAUDE.md `@` imports aren't loading.** `@docs/user-stories.md:` and the other three priority docs are followed by a colon, and `@docs/jobs-to-be-done.md.` by a full stop. That punctuation seems to become part of the path. Only `app-roles.md` was loaded into my context automatically, so the contract docs aren't auto-loaded in a new session. The fix is a space after each path, e.g. `1. @docs/user-stories.md — acceptance criteria`.
2. **Not a git repo yet.** Phase 0 isn't finished. The test freeze depends on git: the `g3-tests-approved` tag and the reviewer's `git log` check.

### Major (spec gaps that would create bugs or flaky tests)

1. **Two tabs can bring cleared history back.** Tab A still holds the old entries in memory, tab B runs Clear all, then tab A calculates and writes everything back. That breaks R-2's privacy rule. §9.3 should say each save reads the current stored list first. Optionally, also listen for the `storage` event.
2. **Restore doesn't clear errors** (technical-spec.md:93). Restoring while errors are showing would display errors and results together, which breaks S-2. Restore also doesn't say to re-validate the stored input. A tampered entry like `amount: "abc"` passes a shape-only type guard.
3. **AC-11.2 depends on the time zone.** §12.1 fixes the clock but sets no `timezoneId`. "20:45" would pass in Karachi and fail on a UTC CI runner. Both should be pinned.
4. **The domain API is undefined, but tests come first.** The test-designer can't create stubs (test-designer.md:21), so its imports will invent the function names and signatures. Those are then frozen at G3. I'd add the signatures to §3.1, e.g. `calculateLoan`, `validateInputs`, `formatMoney`.
5. **AC-8.5 ("Tab stays within the dialog") may not hold with a native `<dialog>`.** As far as I know, browsers let Tab leave a `showModal()` dialog into the browser's own UI. Either add a focus trap or reword the AC to "focus never reaches page content behind the dialog."

### Minor

- **The reviewer's freeze check is wrong** (code-reviewer.md:15). `--since=<commit>` takes a date, not a commit, so it should be `git log -p g3-tests-approved..HEAD -- …`. It also needs to exclude `e2e/review/`, which the reviewer itself writes to.
- **"No dependencies beyond §2" will cause constant stop-and-ask.** Basic tooling isn't listed: `@vitejs/plugin-react`, `@types/react(-dom)`, `@types/node`, `@eslint/js`, `globals`, `eslint-config-prettier`. Pre-approve these at G1.
- `verify` **runs plain** `trace`**, not** `trace --results`**.** So "Implemented means it passed in all 3 browsers" is never actually enforced by the gate.
- **Input borders are only 1.42:1 against the surface.** WCAG 1.4.11 asks for 3:1 for borders that identify a control, and the tests only check text.
- **Clear all leaves the current customer's figures on screen** (AC-12.6). That's defensible, but it sits uneasily with J-7's privacy goal. It's a product call for you.
- **Small copy-deck gaps:** the dialog's accessible name, headings for the form and results cards, and separate keys for the theme options.

- **Possible flaky tests:** AC-10.8's "1 s on CI" doesn't define start and end points. The Pages smoke run straight after deploy may hit propagation lag.
- **Node 24 is still LTS today**, so the spec's condition is met. Node 26 should become Active LTS within days, though, so it's worth a line in the decision log.

## Questions only you can answer (needed at G1)

- **D-05:** 2 dp or whole rupees for PKR? Every reference figure assumes 2 dp, so this has to be settled before Phase 3.
- **D-01 to D-12 are all still Proposed.** CLAUDE.md says not to build on them until you approve them.
- Should Clear all also reset the form?

If you agree, the next step is the Phase 1 prompt from loop-plan.md. I'd run `spec-critic` independently and merge its findings with the ones above into a single G1 list. Fix #1 and #2 first so the session starts with the full context and a repo.

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### 👴 Muhammad Arslan (owner) · 2026-10-17 20:50 PKT

*(had a file open or selected in the editor)*

> 1- should be fixed, 2- we will do at end

🔧 2 tool call(s): Read ×2

### 🤖 Claude · 2026-10-17 20:51 PKT

I read "should be fixed" as you having already made the change. It's on disk, and the import syntax is right now:

- All six paths are followed by a space and then `–`, so the trailing `:/.` that broke them is gone (CLAUDE.md:6-14).
- All six target files exist at those paths.
- I can't prove they load from inside this session, because imports are read when a session starts. In a new session, run `/memory` (or ask Claude to quote a line from `technical-spec.md` without reading it) to confirm.

**There's a side effect.** Whatever you pasted from removed all the Markdown, probably because it came from a rendered preview:

- The `#` headings are gone.
- The hard rules are no longer a list.
- The Commands and Subagents tables are now tab-separated text.
- The backticks and bold are gone, so things like `[AC-x.y] ....`, `*.test.ts(x)` and `e2e/**` no longer stand out as code.

Claude can still follow it, but the hard rules read as one block of prose, which makes them easier to skim past. Do you want me to put the Markdown back as it was in the original, keeping your import fix? Since there's no git yet, I'd show you the diff before writing.

**On #2 (git at the end):** fine by me. Just note that two things rely on git history existing by G3. One is the `g3-tests-approved` tag behind the test freeze; the other is the reviewer's check that no test changed after G3. CLAUDE.md's rule about small, conventional commits also assumes a repo. If git only arrives at the end, that evidence won't exist. A local `git init` before Phase 3, without pushing to GitHub, would cover it.

### 👴 Muhammad Arslan (owner) · 2026-10-17 20:53 PKT

> what questions do you have for me right now?

### 🤖 Claude · 2026-10-17 20:53 PKT

Only the first question blocks anything right now. The others can be answered at G1, but they shape what gets built, so here they are.

**Now**

1. **CLAUDE.md formatting:** should I put back the Markdown that the paste removed (headings, list, tables, code formatting), keeping your import fix? I'll show you the diff first. *My recommendation: yes.*
2. **Is this session Phase 1?** The loop plan has you type the Phase 1 prompt yourself, and it expects one transcript per phase. Should this session be exported as a pre-read, or treated as the start of Phase 1? *My recommendation: export it as a pre-read and start Phase 1 in a fresh session, which also lets you check that the imports load.*

**Product decisions (yours alone)**

1. **D-05:** keep 2 decimals for PKR, or show whole rupees? Every reference figure assumes 2 dp. *My recommendation: keep 2 dp.*
2. **Clear all and the screen:** should Clear all also reset the form? Today AC-12.6 keeps the current customer's figures on screen, which works against J-7 (privacy). *My recommendation: keep AC-12.6 as it is, because Reset already does that job, and add one line to the dialog text. That's a copy-deck change you'd approve.*
3. **The focus rule in AC-8.5:** should I add a JS focus trap to the dialog, or will you reword the AC to "focus never reaches the page behind the dialog"? *My recommendation: reword it, since the browser's native behaviour is the more accessible one.*

**Spec changes I'd like your go-ahead on**

1. **Two-tab history:** each save re-reads stored history before writing, so cleared entries can't come back.
2. **Restore:** it clears field errors, and stored entries go through full validation, not just a shape check.
3. **Test time zone:** pin the E2E tests to `Asia/Karachi` so AC-11.2 gives the same result locally and in CI.
4. **Domain API:** I'd add function signatures to §3.1 before Phase 3, so the test-designer doesn't invent the API that then gets frozen.
5. **Tooling dependencies:** pre-approve these so the build doesn't stop to ask for each one: `@vitejs/plugin-react`, `@types/react`, `@types/react-dom`, `@types/node`, `@eslint/js`, `globals`, `eslint-config-prettier`.
6. `verify`**:** run `trace --results`, so the "Implemented means it passed in 3 browsers" rule is actually enforced.
7. **Input borders:** darken `--border` to reach 3:1, or give inputs a filled background instead.

**Process**

1. **Git:** is a local `git init` before Phase 3 acceptable, with the GitHub push still at the end? Without it there's no G3 tag and no evidence that the tests stayed frozen.

If it's easier, answer just 1 and 2 now and leave the rest for the G1 list.

### 👴 Muhammad Arslan (owner) · 2026-10-17 20:57 PKT

1-CLAUDE.md formatting: yes. 2-Is this session Phase 1? No this is planning phase Product decisions (yours alone)

1. D-05: keep 2 decimals for PKR, or show whole rupees? Every reference figure assumes 2 dp. keep 2 dp.
2. Clear all and the screen: should Clear all also reset the form? Today AC-12.6 keeps the current customer's figures on screen, which works against J-7 (privacy).keep AC-12.6 as it is, because Reset already does that job, and add one line to the dialog text. That's a copy-deck change you'd approve.
3. The focus rule in AC-8.5: should I add a JS focus trap to the dialog, or will you reword the AC to "focus never reaches the page behind the dialog"? reword it, since the browser's native behaviour is the more accessible one.

Spec changes I'd like your go-ahead on

1. Two-tab history: each save re-reads stored history before writing, so cleared entries can't come back. Approved
2. Restore: it clears field errors, and stored entries go through full validation, not just a shape check. Approved
3. Test time zone: pin the E2E tests to Asia/Karachi so AC-11.2 gives the same result locally and in CI. Approved
4. Domain API: I'd add function signatures to §3.1 before Phase 3, so the test-designer doesn't invent the API that then gets frozen. Approved
5. Tooling dependencies: pre-approve these so the build doesn't stop to ask for each one: @vitejs/plugin-react, @types/react, @types/react-dom, @types/node, @eslint/js, globals, eslint-config-prettier. APproved
6. verify: run trace --results, so the "Implemented means it passed in 3 browsers" rule is actually enforced. APproved
7. Input borders: darken --border to reach 3:1, or give inputs a filled background instead. Approved

Process

1. Git: is a local git init before Phase 3 acceptable, with the GitHub push still at the end? Without it there's no G3 tag and no evidence that the tests stayed frozen. Let's create a repo then.

Right you are going to come up with solid plan, You will implement everything in phases story by story.

### 🤖 Claude · 2026-10-17 20:58 PKT

I've already read every file in this folder, and there's no code to explore, so I'll go straight to the planning facts I'm missing. I'll check `gh` and the git identity, and pick border colours that reach 3:1.

🔧 2 tool call(s): Bash, AskUserQuestion

### 🤖 Claude · 2026-10-17 20:59 PKT

Now I'll write the draft plan.

🔧 1 tool call(s): Write

### 🤖 Claude · 2026-10-17 21:00 PKT

The draft is written. Before I show it to you, an independent planner will check it against the docs, because the domain API and spec changes get frozen at G3.

🔧 4 tool call(s): Agent, Write, ToolSearch, ExitPlanMode

### 🤖 Claude · 2026-10-17 21:09 PKT

The plan is approved, so I'm starting with P0: version control, with commit 1 holding the pack exactly as received.

🔧 2 tool call(s): Bash ×2

### 🤖 Claude · 2026-10-17 21:09 PKT

Commit 1 is in (`6d2986d`). Next is P1: restoring the CLAUDE.md formatting, written to `CLAUDE.md.next` so you can review the diff first.

🔧 8 tool call(s): Write, Bash, Read ×6

### 🤖 Claude · 2026-10-17 21:10 PKT

I've read all the files. Now the spec edits, section by section.

🔧 14 tool call(s): Edit ×14

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### 👷 Muhammad Arslan (owner) · 2026-10-17 21:11 PKT *(sent while Claude was working)*

> one other thing, keep all decision logs

🔧 2 tool call(s): Edit ×2

### 🤖 Claude · 2026-10-17 21:11 PKT

Now the decision log: append-only, with D-05's original text kept.

🔧 5 tool call(s): Edit ×5

### 🤖 Claude · 2026-10-17 21:12 PKT

Next: the two agent files, the loop plan, and the decision-log rule in `CLAUDE.md.next`.

🔧 4 tool call(s): Edit ×4

### 🤖 Claude · 2026-10-17 21:12 PKT

Next is the Phase 6 order change in the loop plan.

🔧 5 tool call(s): Grep, Edit ×2, Write ×2

### 🤖 Claude · 2026-10-17 21:12 PKT

All edits are in. Now the integrity check: the only lines removed anywhere should be the ones the plan intended to change.

🔧 2 tool call(s): Bash ×2

### 🤖 Claude · 2026-10-17 21:13 PKT

The planning-phase edits are written but not committed. The plan has me show you two diffs before commit 2.

**Done so far**

- **Git repo created** on `main`, with your identity set for this repo only. Commit 1 (`6d2986d`) is the pack exactly as received, plus a `.gitignore`. One extra I added that wasn't in the plan: `core.autocrlf=false`, so files don't get line-ending changes on Windows. It's recorded in D-22.
- **technical-spec.md:** all the approved changes are in.

  - The domain API is now §3.1.1, including the throwing-stub rule.
  - §3.2 (Restore), §2, §7, §8.3, §8.4, §9.2, §9.3, §12.1, §12.4, §13 and §15 are updated as planned.
- **decision-log.md:** nothing was removed except D-05's status cell. D-05 keeps its original proposal and sub-question, and gains a dated **Decision** line.
  - D-13 to D-24 are added. D-22 covers git, D-23 the freeze scope, and D-24 the rule that the log is append-only.
  - D-01 to D-04 and D-06 to D-12 stay **Proposed** until G1.
- **Agent files:** the code-reviewer's freeze check now uses a command that works, and it runs group-scoped checks until G4.6. The test-designer now knows about the stubs and may write `test/**`.
- **loop-plan.md:** the Phase 0 and G2 notes about the remote, and Phase 6 now runs `trace --results` before the README and push.
- **"Keep all decision logs":** this is now a hard rule in `CLAUDE.md.next` and D-24, and saved to memory for future sessions.

**Please approve before I commit**

1. **The AC text above** (AC-8.5 and AC-12.2). These are the only edits to `user-stories.md`, and no Status changed.
2. **The CLAUDE.md swap.** The Markdown is restored and your six import lines are kept as they are. The content changes are three added hard rules (import `Decimal` only from `money.ts`, the pre-approved tooling pointer, and the decision-log rule), the freeze rule pointing to §12.4, and `test/**` in the test-designer row.

When you reply OK, I'll:

- move `CLAUDE.md.next` into place;
- make commit 2, `docs: apply planning-phase decisions (D-05, D-13 to D-24)`;
- create the `v1` branch;
- show you `git log --stat`.

After that, run `/export transcripts/session-00b-planning.md` and start Phase 1 in a new session. Running `/memory` there will confirm that all six docs load.

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### 👱 Muhammad Arslan (owner) · 2026-10-17 21:15 PKT

> ok

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-17 21:16 PKT

The planning phase is finished and committed.

- **Commits on `main`:** `6d2986d` is the pack exactly as received, and `b56ecb8` holds the planning decisions (D-05 approved, D-13 to D-24 added, plus the AC-8.5 and AC-12.2 text). Your restored CLAUDE.md is part of the second commit.
- **Branch:** the `v1` branch is created and checked out. Phases 1–5 happen there.
- **Status:** nothing is left uncommitted, and commits are authored as `arslan8954@gmail.com`.

**Your next steps**

1. Run `/export transcripts/session-00b-planning.md` in this session. The file will show as untracked until it's committed at the start of Phase 1.

2. Start a new session in this folder and run `/memory` to check that all six docs load. If they don't, I'll fix it before anything else.
3. Paste the Phase 1 prompt from loop-plan.md. In that session I explain the product back to you and run `spec-critic` on the updated docs. Then I give you one merged list, and stop at G1 for your decisions on D-01–D-04 and D-06–D-12.

**Still open for later:** the public GitHub repo. Please create it on github.com by G4.6 at the latest, so CI can run the AC-10.8 timing check.

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### 👴 Muhammad Arslan (owner) · 2026-10-17 21:20 PKT

*(had a file open or selected in the editor)*

> okay start phase 1

### 🤖 Claude · 2026-10-17 21:20 PKT

Starting Phase 1 here. Phase 1 will be part of this same session, so the `session-00b-planning` export will include it. If you'd rather keep a separate file, export once now and again at G1. One check can't happen here: you'll only see whether the imports load when you next start a fresh session.

I'm starting `spec-critic` on the updated docs in the background and doing my own pass while it runs.

🔧 1 tool call(s): Agent

### 🤖 Claude · 2026-10-17 21:20 PKT

`spec-critic` is still running. While it works, I've noted the points from my own pass that it won't raise: they come from earlier in this session or from things only I know about the environment. When its report arrives, I'll merge everything into one list ordered by severity, with a proposed fix for each, alongside my explanation of the product back to you. Then I'll stop at G1.

### 🤖 Claude · 2026-10-17 21:26 PKT

The critic returned 38 findings. Before passing them on, I'll check the two I have reason to doubt: #12 (CLAUDE.md not updated) and #8 (precision of `r`). For #8, the argument I gave you earlier doesn't hold at the largest balances, so I need to re-test.

🔧 2 tool call(s): Grep, Bash

### 🤖 Claude · 2026-10-17 21:27 PKT

# Gate G1: spec challenge

## 1. My understanding of the product

- **Who it's for:** three kinds of people use the same screens.
  - **R-1 Prospective Borrower:** not a finance expert, often on a phone, asking "can I afford this, and what will it really cost?"
  - **R-2 Loan Advisor:** quotes customers many times a day and needs speed, keyboard entry, exact and explainable figures, saved quotes, and privacy between customers.
  - **R-3 Existing Borrower:** already repaying and wants to see where each payment goes and what is still owed.
- **v1 includes (S-1 to S-13):**

- the monthly payment, total interest and total amount payable, with strict validation and messages;
- results that disappear as soon as an input changes;
- a phone layout, light/dark themes, and PKR/USD symbols with no conversion;
- full keyboard use and a one-step Reset;
- the month-by-month schedule;
- up to 20 saved calculations in this browser, with Restore, Delete and Clear all.

It is client-only, with no network calls after load.

- **v1 excludes:** S-14 to S-19 (comparing offers, a maximum loan from a budget, copy, export, prepayments, yearly summaries). It also excludes fees, flat-rate or variable loans, payment dates, currency conversion, accounts and other languages.
- **How the schedule works:** this is a reducing-balance loan with `r = rate/1200`. Each month:
  - interest = the opening balance × r, rounded half-up to 2 dp;
  - the payment is recomputed as the EMI of the **remaining** balance over the remaining months, then rounded;
  - principal = payment − interest;
  - the last month pays off whatever balance is left.

The headline figure is month 1's payment. Every month is within 0.01 of it, and the balance ends at exactly 0.00. The textbook "fixed payment, last one adjusts" method is ruled out because it fails within the accepted inputs.

## 2. What I checked (evidence)

- `spec-critic` read every doc, CLAUDE.md, the loop plan and the 3 agents, and changed nothing. It returned **38 findings** (17 Major, 21 Minor) and 9 questions.
- **I re-checked 2 of them myself:**
  - **#12 is wrong.** CLAUDE.md line 28 already says "or anything else in the freeze scope of spec §12.4", so I've dropped it.
  - **#8 (rounding of `r`): I need to correct something I told you earlier.** I said precision was "safe" with a short proof, but that proof doesn't hold at the largest balances. I've now tested **23,234** exact half-cent tie cases, covering every rate from 0.01% to 50% at both the smallest and largest balances, and found **0 mismatches** against exact arithmetic. The results are correct, but they rest on a test rather than a proof. So I'd change the spec to `round2(balance × rate / 1200)` (multiply first). That form is exact by construction and gives identical results. I'm downgrading #8 to Minor.

## 3. Decisions only you can make (my recommendation first)

1. **Commas in the amount (#3, D-10).** Today `1000,50` becomes **100050** and `1,5` becomes **15**. Phones with a decimal-comma keypad can produce those, and the user would then get a confident figure for the wrong loan. *Recommendation:* accept commas only in valid western grouping (`1,000,000`) or lakh grouping (`10,00,000`), only in the integer part. Anything else gets `Loan amount must be a number.` Approve D-10 with that change.
2. **Idle text next to errors (#2).** After a failed Calculate, the state is `idle`, so the results card would show `Enter your loan details and select Calculate.` But AC-2.12 says "only the error message is shown". *Recommendation:* show the idle text, and reword AC-2.1, AC-2.12 and AC-9.1 to "results and schedule are removed; the results area shows the idle text".
3. **When delete or Clear all fails to write (#4).** Today the UI could report "cleared" while the data is still on the device, which breaks R-2's privacy guarantee. *Recommendation:* keep the entries visible and show

new copy: `Couldn't remove saved calculations from this browser. To be sure, clear this site's data in your browser settings.`

4. **Duplicate accessible names (#6).** Two entries saved in the same minute both get the name `Restore calculation from 17 Oct 2026, 20:45`. AC-8.6 creates exactly that situation. *Recommendation:* use `Restore {line1}, saved {datetime}`, and the same pattern for Delete.
5. **Focus after the dialog (#7).** *Recommendation:* Cancel and Escape send focus back to **Clear all**. After Clear, Clear all becomes disabled, so focus goes to the **Saved calculations** heading. Focus is set in code; React's `autoFocus` isn't used, because it fires when the component mounts, not when the dialog opens.
6. **The "3 browsers" rule for domain-only ACs (#1).** AC-3.6, AC-10.4, AC-10.5 and AC-13.3 are proven only in Vitest, which never runs in a browser. As written, `trace --results` could therefore never pass them. *Recommendation:* every AC needs at least one passing test, and any Playwright test must pass in all 3 browsers.
7. **AC-10.8 timing (#14).** *Recommendation:* measure in the page from the Calculate click until the 480th row is in the DOM, ≤ 1,000 ms, in all 3 browsers. Check it locally from G4.3, with CI confirming at G4.6.
8. **Product fit (Q3).** Many Pakistani car and Islamic-finance quotes are flat-rate or markup-based, so the app's figures won't match them. *Recommendation:* no scope change, but state it clearly in the README's assumptions.
9. **Still-Proposed decisions.** The critic found no problems with D-01, D-03, D-04, D-06, D-07, D-08, D-09, D-11 or D-12.
10. D-10 depends on your answer to Q1.
11. D-02's "proven correct across the whole range" is an over-claim. I'd add a dated note saying "verified over ~1,700 cases", rather than rewriting it (D-24).

*Recommendation:* approve all of them, with those two adjustments.

## 4. Fixes I'd apply without a product decision (approve as a batch)

- **ACs (your approval needed, since they're AC text):**
  - AC-5.3 uses L-B, because L-A's 12 rows may fit without scrolling (#15).
  - AC-11.10 is split into cases: not JSON, not an array, a mix of good and bad entries (the good ones are kept), `[null]`, and a tampered payment (#10).
  - AC-8.1 is split into the order on an empty page and the order with results and entries showing (#18).
  - Observable wording for AC-2.15 ("responsive") and AC-5.2 ("fully visible") (#19).
  - AC-11.2 states the time zone and the three-line structure (#24).
  - One AC for the same input in a different currency, which adds an entry (#23).
- **Spec:**
  - Test seeding: saved entries use 0% loans (for example, Rs 12,000 over 12 months is Rs 1,000.00 a month, which is known by hand), or are created by driving the UI. That fixes #11 without computing expected values.
  - Biased generators, plus the two §5.3 failure loans as `[SPEC-§5.3]` tests (#9).
  - Validate settings, with fallback to the defaults (#17).
  - Handle `key === null` in the `storage` event (#25).
  - A getter-throws stub for Safari's blocked storage (#26).
  - Guard the dialog in jsdom (#27).
  - Use one Decimal configuration everywhere: `clone` (#28).
  - Drop the "100%" claim, add `--max-warnings 0` and a no-cookies check, and word NFR-1 by engine (#29).
  - Copy keys for the aria-labels (#30).

    - `type="button"` on Reset and the dialog buttons (#31).
    - The negative-only tests allowed to pass in Phase 3 are listed (#32).
    - Input edge cases as a `[SPEC-§4.2]` table (#20).
    - Stale-state and Restore edge cases (#21, #22).
    - Formatter input contract (#38).
- **Process:**
    - The Phase 2 prompt names the stubs, TZ config, `test/setup.ts` and `--pass-with-no-tests`, and exits on individual scripts rather than `verify` (#13).
    - The reviewer gets the Edit tool, so findings aren't overwritten (#33).
    - The developer owns the configs, and the test-designer's rules include component tests (#34).
- **Smaller notes:**
    - D-24 note: updating the Status column in the index is allowed (#35).
    - Node 24 is pinned deliberately (#37).
    - J-7 vs Reset (#5): I'd leave the mappings and rely on D-21. Changing them means editing `jobs-to-be-done.md`, which only you can do.
    - My own additions:
        - Local `verify` runs Windows WebKit, so the 3-browser evidence that counts comes from CI on Linux.
        - Running Phase 1 in this session gets recorded as **D-25**.

### 5. Not done

I haven't edited any docs or committed anything, because Phase 1 stops here. Answer Q1–Q9 (just "as recommended" is fine) and approve the batch in §4. I'll then apply everything, record every decision in the log, and commit `docs(G1): ...`. Phase 2 doesn't start until you say so.

---

#### 👴 Muhammad Arslan (owner) · 2026-10-17 21:32 PKT

> seems good, let's move to phase 2

#### 🤖 Claude · 2026-10-17 21:34 PKT

Taking "seems good" as approval of every G1 recommendation, Q1–Q9 plus the §4 batch. First I'll write those into the docs and the decision log and commit them, which closes G1. Then Phase 2.

**G1 edits: technical spec**

🔧 34 tool call(s): Edit ×34

#### 🤖 Claude · 2026-10-17 21:36 PKT

Next, the AC edits in user-stories.md that the G1 decisions require:

🔧 18 tool call(s): Read ×3, Edit ×15

#### 🤖 Claude · 2026-10-17 21:37 PKT

Now the decision log. I'm adding G1 approvals as dated Decision lines and adding D-25 to D-40. No past text is rewritten; only the Status column changes, which D-24 allows.

🔧 8 tool call(s): Bash, Read, Edit ×6

### Claude · 2026-10-17 21:38 PKT

Next, the process files: loop-plan Phase 2 and 3, and the two agents.

6 tool call(s): Edit ×6

### Claude · 2026-10-17 21:38 PKT

Before committing, I'll check that the decision log lost nothing but the Status cells.

1 tool call(s): Bash

### Claude · 2026-10-17 21:39 PKT

G1 is closed (`ae03c49`), and the decision log lost only the Status cells. **Phase 2: scaffold.** First I'll find out what versions npm currently offers:

2 tool call(s): Bash, Read

### Claude · 2026-10-17 21:42 PKT

These are newer major versions than I know well: TypeScript 7, Vite 8, Vitest 5 and ESLint 10. I'll check the peer-dependency ranges before choosing exact versions.

2 tool call(s): Bash ×2

### Claude · 2026-10-17 21:46 PKT

The choice is TypeScript 6.0.3 and ESLint 9.39.5. Everything else is at its latest version, except `@types/node`, which is pinned to 24.x to match the Node 24 runtime. Writing the project files now.

3 tool call(s): Write, Bash ×2

### Claude · 2026-10-17 21:49 PKT

The configs are next. I'll start with TypeScript, Vite, Vitest and Playwright:

7 tool call(s): Write ×7

### Claude · 2026-10-17 21:49 PKT

Next is the ESLint config. First I'll check how these plugin versions expose their flat configs:

4 tool call(s): Bash, Write ×3

### Claude · 2026-10-17 21:50 PKT

The `money.ts` exemption above also turned off its layer rule. I'll narrow it so only the `decimal.js` import is allowed:

5 tool call(s): Edit, Write ×4

### Claude · 2026-10-17 21:50 PKT

Now the domain layer: `types.ts` and the configured `Decimal` in full, and every other §3.1.1 function as a throwing stub with its exact signature.

7 tool call(s): Write ×7

### Claude · 2026-10-17 21:51 PKT

Now the two scripts, size and traceability, both using Node only:

2 tool call(s): Write ×2

### Claude · 2026-10-17 21:51 PKT

The browsers are downloaded. Next, the two GitHub Actions workflows:

🔧 2 tool call(s): Write ×2

### Claude · 2026-10-17 21:51 PKT

Everything is in place. Now I'll run each `verify` step separately and fix what fails:

🔧 2 tool call(s): Bash ×2

### Claude · 2026-10-17 21:53 PKT

Six of the seven non-trace steps pass. The bundle is 66.2 KB gzipped against the 150 KB limit. Next are `test:e2e` and `trace`:

🔧 10 tool call(s): Bash ×5, Edit ×5

### Claude · 2026-10-17 21:55 PKT

Everything is committed and the tree is clean. I'll do one last run of every step on the committed tree, so the gate evidence matches what's in git:

🔧 1 tool call(s): Bash

### Claude · 2026-10-17 21:56 PKT

## Gate G2: scaffold done

Phase 2 is finished and committed on branch `v1`. The app is an empty shell showing only its title. Every `verify` step passes on its own except `trace`, which is supposed to fail until Phase 3 writes the tests. That's the exit condition we agreed.

**First, G1 is closed** (`ae03c49`). I recorded all your G1 decisions in the log as D-25 to D-40. In the user stories I reworded 13 ACs and added two new ones, AC-2.17 and AC-12.7. No story Status changed. I rejected one critic finding (#12) because it was wrong.

**What Phase 2 contains** (commits `b322938`, `c78f28d`, `90be39d`)

- **Tooling:** Vite, React and strict TypeScript, all at exact pinned versions.
- **Domain layer:** `types.ts` and the configured `Decimal` are written in full. Every other function in the agreed domain API exists with its exact signature and throws `NOT_IMPLEMENTED`. That's what will make Phase 3 tests fail for the right reason.
- **ESLint guards:** the layer boundaries, `decimal.js` only in `money.ts`, and bans on `Intl`, `parseFloat` and `toLocale*`.
- **Test setup:** Vitest (domain and UI projects, Karachi time zone, JSON report), Playwright (3 browsers, time zone, local server, smoke mode), and `test/setup.ts`.
- **Scripts:** `trace.mjs` and `size.mjs`, using Node only.
- **Workflows:** `ci.yml` and `deploy.yml`.

**Evidence** (final run on the committed tree)

| Step | Result |
|---|---|
| lint, format:check, typecheck, test, build, test:e2e | exit 0 |
| size | exit 0, 66.2 KB of 150 KB gzipped |
| trace | exit 1, as expected: 94 of 94 v1 ACs untested (100 ACs in total, 19 stories, 11 jobs, 3 roles) |

`test:e2e` passing with no tests proves nothing by itself, so I also ran throwaway probes and deleted them afterwards:

- **Browsers:** the shell loads and the Karachi time zone applies in Chromium, Firefox and WebKit (3 of 3 passed). The Vitest time-zone probe also passed.
- **ESLint guards:** all 7 deliberate violations were caught.
- **trace:** in a scratch copy of the docs with S-1 set to `J-99`, it failed three separate ways (exit 1).

**What I'm unsure about**

- **D-41, which needs your decision.** The spec asks for "latest stable", but that's impossible here. TypeScript 7 is out, yet `typescript-eslint` only supports TypeScript up to 6.0. ESLint 10 is out, yet `jsx-a11y` only supports ESLint 9. I installed **TypeScript 6.0.3 and ESLint 9.39.5**, and npm flags ESLint 9 as no longer supported. The options:
  - (a) keep this setup;
  - (b) move to ESLint 10 and force `jsx-a11y` onto it with an npm override;
  - (c) drop `jsx-a11y`.

I recommend (a).

- **D-42** lists the scaffold choices for your review. The ones that matter most:
  - `@types/node` is pinned to 24.x to match the runtime;
  - the build type-checks only the app, so a broken test file can't block it;
  - Playwright refuses `.only`, and doesn't retry locally;
  - tests must navigate with `page.goto('./')` so they work on GitHub Pages.
- **CI is untested.** The workflows haven't run because there's no GitHub remote yet.
- **All browser runs so far were on Windows,** where Playwright uses its Windows WebKit build. The 3-browser evidence that counts comes from the Linux CI runs.

**Not done**

- **No GitHub push**, so the CI and Pages checks for G2 wait until you create the repo.
- **Full `verify` not run yet:** it fails at `trace` until Phase 3 by design.

**Your G2 checks**

1. Run `npm ci && npm run dev` and open the page. You should see the title "Loan EMI Calculator".
2. Run `npm run trace`. Then change `J-1` to `J-99` in S-1's row in `docs/user-stories.md`, run it again and confirm it fails, then undo the change.
3. Decide on D-41 (a, b or c) and check the D-42 choices. The configs are frozen at G3, so this is the cheap moment to change them.

Phase 3 (tests first) starts when you approve G2.

---

### 👴 Muhammad Arslan (owner) · 2026-10-17 22:25 PKT

> okay run the app

🔧 2 tool call(s): Bash, Read

### 🤖 Claude · 2026-10-17 22:25 PKT

The server is still starting. I'll check it directly rather than watching the log:

🔧 1 tool call(s): Bash

### Claude · 2026-10-17 22:26 PKT

The server responds. `curl` only sees the HTML shell, though, so I'll take a real browser screenshot to confirm React renders the title:

🔧 2 tool call(s): Bash, Read

### Claude · 2026-10-17 22:26 PKT

The app is running at **http://localhost:5173/**. Open it in your browser.

I loaded it in Chromium to check. It shows the heading "Loan EMI Calculator" with 0 console or page errors.

That's all there is for now, and it's expected. Phase 2 is only the empty shell: no styling, form or results yet. The real UI arrives in Phase 4, after the tests are written in Phase 3.

The dev server keeps running in the background until you ask me to stop it.

Two G2 checks are still yours to do:

1. **Fresh install:** run `npm ci && npm run dev` yourself, which checks the setup from a clean install.
2. **Trace check:** change `J-1` to `J-99` in S-1's row and confirm `npm run trace` fails, then undo the change.

The open decision is **D-41**. I need you to pick (a) keep TypeScript 6 and ESLint 9, (b) ESLint 10 with an override, or (c) drop `jsx-a11y`. I recommend (a). Phase 3 starts when you approve G2.

---

### Muhammad Arslan (owner) · 2026-10-17 22:27 PKT

> ;et

### Claude · 2026-10-17 22:27 PKT

I didn't catch that: ";et" looks like a typo, maybe for "let's..." that got cut off.

Because this is the G2 gate, I won't treat it as approval. Could you tell me what you meant? For example:

- **"approve G2, D-41 (a)"** and I start Phase 3 (tests first);
- a different choice for D-41, (b) or (c);
- something else you want changed first.

The dev server is still running at http://localhost:5173/.

---

### Muhammad Arslan (owner) · 2026-10-17 22:27 PKT

> let's do phase 3

### Claude · 2026-10-17 22:28 PKT

Taking "let's do phase 3" as G2 approval. You didn't pick an option for D-41, so my recommendation (a) stands: TypeScript 6 and ESLint 9. I'll record exactly that in the log; tell me if you meant (b) or (c).

There's one gap to fix before the tests get written: **the spec doesn't define how tests locate each figure on screen**. Without that, the test-designer would invent a DOM structure, and the frozen tests would lock it in. So I'm adding a small test-facing DOM contract, D-43, that uses only roles and accessible names. You can review it at G3.

🔧 8 tool call(s): Edit ×5, Bash, Agent ×2

### Claude · 2026-10-17 22:30 PKT

Both `test-designer` agents are running in the background. This will take a while, because the UI agent has to cover all 94 v1 ACs across 13 story specs, plus the spec-rule tests and the manual checklist.

When they finish, I'll check the work myself rather than relying on their reports:

- every test fails for the right reason (a `NOT_IMPLEMENTED` stub or a missing element), not a fixture, syntax or type error;
- the only tests that pass early are the allowed exceptions (negative-only tests, the time-zone test, the title check);
- expected values are hard-coded from the ACs and never computed by app code;
- `trace` shows 0 untested v1 ACs, and lint, format and typecheck pass on the test files.

Then I'll stop at G3 with the evidence and pick 3 tests for you to read closely. The dev server is still running at http://localhost:5173/ in the meantime.

### Claude · 2026-10-17 22:37 PKT

The domain agent has finished. Before the UI agent reports, I'll check its work myself, so I'm not relying on its report alone:

🔧 2 tool call(s): Bash ×2

### Claude · 2026-10-17 22:37 PKT

I've confirmed the domain side independently. Domain tests: **292 fail and 2 pass, out of 294.**

- **Failure reasons:** every failure is a `NOT_IMPLEMENTED` stub. The 7 "Property failed" lines have `NOT_IMPLEMENTED` as their cause; there are no type, import or fixture errors.
- **The 2 passes** are the expected exceptions: the time-zone test, and the check that `Decimal` is configured with 40 digits and half-up rounding.
- **No skips:** there's no `.skip`, `.only`, `.todo` or `.fails` anywhere.
- **Oracle:** the property tests use only the configured `Decimal` to check results. They never use `round2` or `emi`; the only `round2` match is in a test title.

The UI/E2E agent is still running. When it finishes, I'll check its tests the same way, run `trace` for 0 untested ACs, and then stop at G3.

### Claude · 2026-10-17 23:26 PKT

Both agents are done. I'll verify the UI agent's work myself, starting with the full Playwright run across all 3 browsers: its last fixture edit was never followed by one. That run takes about 24 minutes, so it goes in the background while I do the static checks.

🔧 7 tool call(s): Bash ×3, Grep, Read ×3

### Claude · 2026-10-17 23:27 PKT

Committed: `ada505f` (domain), `41aac8d` (E2E) and `06273a5` (manual checklist). The full three-browser run is still going, about 24 minutes in total. When it reports, I'll confirm the failure reasons and give you the G3 summary.

🔧 5 tool call(s): Read, Bash ×2, Edit ×2

**Claude · 2026-10-17 23:55 PKT**

# Gate G3: tests written, all failing as they should

Every v1 AC now has at least one test (`trace: OK`). Almost all of them fail, for the right reason: the code they test doesn't exist yet. Once you approve, I tag `g3-tests-approved`, and from then on the tests and test configs are frozen (D-12, D-23).

**What was done** (written by two `test-designer` agents; commits `ada505f`, `41aac8d`, `06273a5`, `a8dbab0`)

- **Domain:** 294 Vitest tests in `src/domain/*.test.ts` and `test/`. The property tests use fast-check: 1,000 runs per property, seed 20261017, with generators biased towards the edge cases.
- **E2E:** 149 Playwright tests per browser. There's one spec file per story (S-1 to S-13), plus `spec-rules.spec.ts` for the spec rules and the shared fixtures in `e2e/fixtures.ts`.
- **Manual checklist:** manual-test-checklist.md, one line for each of the 94 v1 ACs.
- `@smoke` is on AC-1.1, 2.1, 7.2 and 11.2.

**Evidence** (from my own runs, not the agents' reports)

| Check | Result |
|---|---|
| Vitest (domain) | 292 failed, 2 passed (294). Every failure is a `NOT_IMPLEMENTED` stub. The 2 passes are the time-zone test and the `Decimal` config check, both allowed. |
| Playwright, 3 browsers | 444 failed, 3 passed (447), 0 flaky, 0 skipped. About 24 minutes while red. |
| Playwright, Chromium detail | All 148 failures are missing elements or checks against absent UI; 0 code, fixture or type errors. The 1 pass is the `<h1>` and `document.title` check, which is allowed. |
| `npm run trace` | `trace: OK`: 0 untested v1 ACs |
| lint, format:check, typecheck | All exit 0 on the whole repo |
| Static audit | No `.skip`, `.only`, `.fixme` or `test.fail`; no `goto('/')`; no app code imported into E2E. Property checks use `Decimal` directly, never `round2` or `emi`. |

**3 tests to read closely** (the most edge cases)

1. loan.property.test.ts:289 and the property tests around it. They check every row of 1,000 generated loans against the schedule invariants.
2. spec-rules.spec.ts:177: Clear all in tab B, then Calculate in tab A. `storage` events are switched off in tab A, so only re-read-before-write (D-13) can make it pass.
3. s08-keyboard.spec.ts:202-252: the dialog focus rules from AC-8.5 and D-30.

**What I'm unsure about: please approve or correct before the freeze**

- **D-43, the DOM contract (spec §8.3.1):** how tests find each element on screen, by role and accessible name.
- **D-44, twelve readings the test-designers made.** These bind once the tests freeze. Check these four:
  - **#4:** what "side by side" means in AC-5.5;
  - **#5:** each history entry shows its three lines as separate lines of text;
  - **#9:** comma handling when normalising the amount;
  - **#12:** the timeouts set in the test fixtures.
- **Two findings from the agents:**

- Playwright's WebKit doesn't pass `storage` events between pages opened with `newPage()`, so the two-tab tests open a popup instead.
- AC-8.1 assumes Tab reaches buttons in WebKit. If Windows WebKit behaves differently, AC-8.1 will fail there, and I'll report it rather than weaken the test.

**Not done**

- No component tests. Everything you can see on screen is tested in E2E, which the spec allows.
- I re-ran only Chromium with full detail. For Firefox and WebKit I have the totals from the full run (444 of 447 failing), not a per-cause breakdown.

**Your G3 checks**

1. Read the 3 tests above against their ACs. Are the expected values hard-coded from the ACs?
2. Spot-check one test per story.
3. Skim the manual checklist.
4. Approve or amend D-43 and D-44.

When you approve, I tag `g3-tests-approved` and start Phase 4, group 1 (domain: S-1, S-2, S-3, S-10, S-13).

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### 👱 Muhammad Arslan (owner) · 2026-10-18 00:22 PKT

> approved

### Claude · 2026-10-18 00:23 PKT

Approved. I'll record D-43 and D-44 as approved, tag the freeze, and then start **Phase 4, group 1 (domain)**.

🔧 8 tool call(s): Bash ×4, Read, Edit ×3

### Claude · 2026-10-18 00:23 PKT

The tests match the spec exactly. Writing the four domain modules:

🔧 5 tool call(s): Write ×5

### Claude · 2026-10-18 00:24 PKT

Now I'll run the domain tests against the real code:

🔧 4 tool call(s): Bash ×3, Agent

### Claude · 2026-10-18 00:25 PKT

The `code-reviewer` is running on group 1. When it reports, I'll fix every Blocker and Major it finds without touching any frozen test, then stop at mini-gate **G4.1** with the evidence.

### Claude · 2026-10-18 00:31 PKT

The reviewer reports **0 blocking findings**, and its checks agree with mine.

It raised three Minors:

- **The tenure comment:** I'll add it.
- **The no-op probe in `e2e/review/`:** that folder belongs to the reviewer, so I'll ask it to delete its own file rather than editing there myself.
- **Year padding below 1000 in dates:** I'll leave it. Stored entries must have a year from 2000 to 9999 (§9.2), so it can't happen.

🔧 3 tool call(s): ToolSearch, Edit, SendMessage

### Claude · 2026-10-18 00:31 PKT

While the reviewer cleans up its probe, I'll record how the Minors are handled (D-45), as D-24 requires:

🔧 2 tool call(s): Edit ×2

### Claude · 2026-10-18 00:31 PKT

The reviewer is still working on the deletion. Once it confirms, I'll check the file is gone, commit the comment fix, the findings file and D-45, and then give you the G4.1 summary.

🔧 0 tool call(s):

### Claude · 2026-10-18 00:31 PKT

The reviewer went idle but didn't confirm, so I'll check the files directly:

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 00:32 PKT

The probe is deleted and the reviewer has noted it in the findings file. I'll rerun the domain tests and lint after the comment change, then commit:

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 00:32 PKT

# Mini-gate G4.1: domain layer done

Group 1 (S-1, S-2, S-3, S-10, S-13 at domain level) is implemented and independently reviewed. **All 294 domain tests pass, and the reviewer found 0 blocking issues.** The commits are `9db17a8`..`47e14e4`, plus `f4002cb` and `4fa3508`.

**What was built**

- `money.ts`: half-up rounding to 2 dp, and `Rs` / `$` formatting with 3-digit grouping and no exponent notation, whatever the size.
- `parse.ts`: trimming, the single trailing `%`, and strict comma rules. Only valid western (`1,000,000`) or lakh (`10,00,000`) grouping is accepted (D-26).
- `validate.ts`: the §4.2 checks in order, with exact error codes. Comparisons use `Decimal`, so 40-digit input is handled correctly, and `-0` is normalised to `0`.
- `loan.ts`: EMI and the re-amortising schedule exactly as §5.2 sets out, with interest computed multiply-first (D-37).
- `format.ts`: the rate, tenure, date-time and canonical-input text.

**Evidence**

| Check | Result |
|---|---|
| `vitest --project domain` | **294 of 294 passed** (18.5 s), including every property test at 1,000 runs, seed 20261017, with no counterexample |
| lint, format:check, typecheck, build, size | All exit 0; bundle 66.2 KB of 150 KB gzipped |
| Freeze check since `g3-tests-approved` | No changes in the frozen scope |
| `code-reviewer` | **0 blocking findings, 3 Minor** (below) |

The reviewer also tried hostile input: `+5`, `1e6`, `Infinity`, `0x10`, Arabic-Indic and full-width digits, `1 000`, a 10,000-character string, and `-0`. Every one was handled correctly, and nothing threw. L-MAX calculates in 16 ms and ends at exactly 0.00.

**The Minors (D-45, for you to review)**

1. A comment now explains why converting tenure to a plain number is safe. Fixed.
2. The reviewer's own probe file asserted nothing, but would have run in all three browsers. It deleted the file at my request.
3. Dates don't zero-pad years below 1000. **Won't fix:** that can't happen, because stored entries must have a year from 2000 to 9999.

**Not done**

- Every E2E test is still red, because there's no UI yet; that's group 2.
- The reviewer's UI-level hunting (double Calculate, two tabs, zoom, the back button) waits until then too.

**Your check:** there's nothing to click yet, so the most useful thing is to read review-findings.md and approve D-45. Then say go for **group 2 (core UI: S-1, S-2, S-3, S-4, S-9)**: the form, results, errors, stale state and Reset.

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### 👱 Muhammad Arslan (owner) · 2026-10-18 00:39 PKT

> next step?

### 🤖 Claude · 2026-10-18 00:39 PKT

The next step is **Phase 4, group 2: the core UI**. It covers S-1, S-2, S-3, S-4 and S-9, which makes the calculator usable:

- the loan form (amount, rate, tenure and unit, with Calculate and Reset), using the full copy deck from §7;
- the results card: Monthly payment, Total interest, Total amount payable and the rounding note;
- error messages under each field, with focus moving to the first invalid field;
- results disappearing as soon as an input changes, with the stale message (S-4);
- Reset (S-9), and the error boundary.

I'll keep going until those stories' E2E tests pass in Chromium, Firefox and WebKit, with lint and typecheck clean. Then `code-reviewer` does an independent pass, and I fix any Blockers or Majors it finds. After that I stop at **G4.2**, and you'll be able to try the calculator in the browser yourself.

Before I start, I need a clear go-ahead from you. "Approve G4.1" covers D-45, the handling of the three Minors. If you want any of those Minors handled differently, say so instead.

---

### 👱 Muhammad Arslan (owner) · 2026-10-18 00:39 PKT

> Can you please restart the app?

🔧 2 tool call(s): ToolSearch, TaskStop

### 🤖 Claude · 2026-10-18 00:39 PKT

I've stopped the old server. On Windows the Vite process can outlive its npm parent, so I'll make sure port 5173 is actually free before starting again:

🔧 1 tool call(s): PowerShell

### 🤖 Claude · 2026-10-18 00:40 PKT

Something is still listening on 5173. Before I stop it, I'll confirm it's our Vite server and not another program:

🔧 4 tool call(s): PowerShell ×2, Bash ×2

### 🤖 Claude · 2026-10-18 00:41 PKT

I've restarted the app. It's running at **http://localhost:5173/**.

On Windows the old Vite process kept running after I stopped its parent, so I checked it was our dev server and stopped it too. Then I started a fresh one on the same port.

It still shows only the "Loan EMI Calculator" title. Group 1 built the calculation code behind the screen, but nothing on the page uses it yet. The form, results and errors come in group 2.

I'm waiting for your "approve G4.1" (it covers D-45) before I start group 2.

---

### 👱 Muhammad Arslan (owner) · 2026-10-18 00:41 PKT

> approved

### 🤖 Claude · 2026-10-18 00:41 PKT

G4.1 is approved. I'll record that, then start **group 2 (core UI)**. First I'll read the frozen E2E specs and fixtures for S-1, S-2, S-3, S-4 and S-9, because they're the contract this UI has to meet.

🔧 2 tool call(s): Bash ×2

### 🤖 Claude · 2026-10-18 00:42 PKT

I've found what looks like an **error in a frozen test**. Before I stop to raise it, I'll read the last group 2 spec (S-2), so any other problems come to you in one batch.

🔧 2 tool call(s): Bash, AskUserQuestion

### 🤖 Claude · 2026-10-18 00:44 PKT

Thanks. Here's the order:

1. Record your decision as D-46.
2. Have the test-designer make exactly that two-line change, in the background.
3. Build the group 2 UI meanwhile.

🔧 3 tool call(s): Edit ×2, Agent

### 🤖 Claude · 2026-10-18 00:45 PKT

While that runs, I'll build group 2. I'm starting with the full copy deck in `copy.ts`, the only place user-visible text lives (spec §7), then the reducer state machine from §3.2.

🔧 2 tool call(s): Write ×2

### 🤖 Claude · 2026-10-18 00:45 PKT

The test fix came back. I'll check the diff myself before committing it, separately from the UI work:

🔧 11 tool call(s): Bash, Write ×10

### Claude · 2026-10-18 00:47 PKT

Next, the stylesheets. `theme.css` holds the colour tokens; for now only the light theme's (group 4 adds dark mode). `app.css` holds the layout. Then main.tsx imports them:

10 tool call(s): Write ×3, Bash ×3, Read ×2, Edit ×2

### Claude · 2026-10-18 00:48 PKT

Lint and typecheck are clean. Next, the group 2 specs in Chromium first, for a fast loop:

15 tool call(s): Bash ×7, PowerShell ×2, AskUserQuestion, Edit ×4, Agent

### Claude · 2026-10-18 00:55 PKT

The diff is exactly the one line. Committing it with D-48 and the spec row, separately from the UI work:

2 tool call(s): Bash ×2

### Claude · 2026-10-18 00:56 PKT

**46 of 48 pass in Chromium.** The 2 failures are exactly the expected cross-group ones: AC-4.4 and AC-9.2 both stop at the missing Theme select, which comes in group 4. Next, Firefox and WebKit:

4 tool call(s): Bash ×3, Edit

### Claude · 2026-10-18 00:59 PKT

Committed in six commits, `9f2c633`..`8c6590a`, and the working tree is clean. Now the independent review of group 2. With a real UI in place, the reviewer can also try hostile interactions:

1 tool call(s): Agent

### Claude · 2026-10-18 01:00 PKT

The `code-reviewer` is running on group 2.

The calculator now works in the dev server at http://localhost:5173/. Vite reloads code changes automatically, so you can try it while the review runs. Enter an amount, rate and tenure and press Calculate. You'll get the summary, the rounding note where it applies, and the full schedule. Invalid input shows messages under each field.

When the review finishes, I'll fix any Blockers or Majors and stop at **G4.2**.

### Claude · 2026-10-18 01:07 PKT

The reviewer confirms 46 of 48 per browser and **0 blocking findings**. I'll fix one Minor now: M-2, linking the currency note to its select, a small accessibility gain. M-5 belongs to group 6 (layout) and M-6 to group 4 (dark theme), so I'll handle them there. M-1 is your decision.

6 tool call(s): Edit ×4, Bash ×2

### Claude · 2026-10-18 01:10 PKT

## Mini-gate G4.2: core UI done

Group 2 is built and has been independently reviewed. **Every test this group can pass now passes in Chromium, Firefox and WebKit, and the reviewer found 0 blocking issues.** The two that still fail need features from later groups. The commits are `9f2c633`..`8c6590a`, plus `b49e75b` and `58f6bcc`.

**What was built**

- **Copy:** the full copy deck in `copy.ts`. All on-screen text comes from there.

- **State:** the reducer follows spec §3.2: edits mark shown results as stale, a failed Calculate clears the results, and Reset empties the form. Changing the unit clears the tenure error.
- **Form:**
  - text inputs with the right keypad hints;
  - error messages under each field, linked to it and announced politely;
  - focus moves to the first invalid field;
  - Enter submits, and Reset moves focus back to Loan amount.
- **Results:** the summary, the rounding note, the schedule table in a focusable scroll area, and the idle and stale messages.
- **Also built:** the error boundary, the currency select (not yet remembered after a reload), and basic light-theme styling.

**Evidence**

| Check | Result |
|---|---|
| S-1, S-2, S-3, S-4, S-9 E2E | **46 of 48 per browser**, in all 3 browsers (`CI=1`). The reviewer confirmed the same. |
| The 2 failures | AC-4.4 and AC-9.2 both stop at the missing **Theme** select (group 4). AC-9.2 also needs saved calculations (group 5). |
| Bonus | S-7's AC-7.1, 7.2 and 7.4 already pass in all 3 browsers. |
| Vitest, lint, format, typecheck | 294 of 294 tests passed; all other checks exit 0 |
| Bundle | 82.7 KB of 150 KB gzipped |
| Freeze check | Only the two changes you approved (D-46, D-48) |
| `code-reviewer` | **0 blocking**, 6 Minor. Its hostile interactions included 10,000-character input, a double Calculate with repeated Enter, the back button and 200% zoom at 360 px. All passed. |

**Two test problems found and fixed with your approval**

- **D-46:** the AC-9.2 test expected `Rs` for entries saved in USD. Two strings were changed.
- **D-48:** my D-43 DOM contract specified a locator that Playwright can never match. The `summary()` helper now uses `getByLabel`; one line was changed. That was my error.

**What I'm unsure about**

- **The stale preview server:** a leftover `vite preview` server once served an old build, and Playwright silently reused it. I now run E2E with `CI=1` to force a fresh build. The permanent fix is one line in the frozen `playwright.config.ts`: `reuseExistingServer: false`. Do you want that change?
- **D-49 and D-50:** my implementation choices for group 2 and how I handled the reviewer's Minors. They're in the decision log for you to approve.

**Try it:** http://localhost:5173/

- Calculate L-A (`1000000`, `12`, `1` Years) and check for **Rs 88,848.79** and the 12-row schedule.
- Then change any value and see the results hide.
- Try `1,,000`, an empty Calculate and Reset.
- Try to break it with something the ACs don't cover.

Reply with "approve G4.2" and your answer on `reuseExistingServer`. Then I'll start **group 3 (schedule UI: S-10, S-13)** and, since the table already exists, verify it against its ACs.

### 👴 Muhammad Arslan (owner) · 2026-10-18 08:20 PKT

> okay let's move on

### 🤖 Claude · 2026-10-18 08:20 PKT

Taking that as G4.2 approved, covering D-49 and D-50. You didn't answer the `reuseExistingServer` question, so I'll leave the frozen config alone and keep using `CI=1`; I'll record it that way. **Group 3 (schedule: S-10, S-13).** The table already exists, so this group starts with a full three-browser run of its specs.

🔧 6 tool call(s): Bash ×2, Agent, Read, Edit ×2

### 🤖 Claude · 2026-10-18 08:22 PKT

I've recorded the G4.2 approvals. The `code-reviewer` is still checking group 3: re-running the schedule tests, measuring how much headroom the 1-second L-MAX timing has, and trying edge cases like switching currency while scrolled down. When it reports, I'll record the outcome and stop at **G4.3**.

### 🤖 Claude · 2026-10-18 08:25 PKT

The reviewer confirms everything: **0 blocking findings**, and the timing has plenty of headroom. Its worst L-MAX time was 150 ms against the 1-second limit.

There are two Minors, and I'm not changing anything for either:

- **G43-1:** the table redraws on every app-state change. But while results are shown, the only change that can happen is switching currency, and that has to redraw the headers anyway; typing hides the table first.
- **G43-2:** a note that group 3 had no code of its own.

I'll record that as D-51 and commit.

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 08:26 PKT

## Mini-gate G4.3: schedule verified

Group 3 (S-10, S-13) is done, and it **needed no new code**. The schedule built in group 2 already passes every test for these stories, and the independent reviewer confirmed it.

| Check | Result |
|---|---|
| S-10 and S-13 E2E | **54 of 54** passed, all 3 browsers (`CI=1`) |
| Other schedule tests (region and table names, sticky header, arrow and Page keys, 360 px with L-MAX) | **15 of 15** passed |
| AC-10.8 (L-MAX, 480 rows in under 1 s) | Reviewer's worst case: **150 ms** in Firefox, about 6.7× headroom. To be re-measured on real CI at G4.6. |
| Freeze check | Only your two approved changes (D-46, D-48) |
| `code-reviewer` | **0 blocking**, 2 Minor (no change for either; see D-51) |

The commit is `5b68f29`, which contains the review findings and D-51.

**Try it** at http://localhost:5173/:

- Calculate L-A and check that row 12 reads `12 | 88,848.78 | 879.69 | 87,969.09 | 0.00`.

- Try L-MAX (`1000000000`, `50`, `40` Years). Click into the schedule and use the arrow keys, Page Up/Down, Home and End.
- Switch to USD: the column headers change to `($)` and the numbers stay the same.

Reply "approve G4.3" (it covers D-51) to move on to **group 4 (settings: S-6 theme, S-7 currency)**. That covers dark mode, the Theme select, both settings remembered across reloads (and kept for the visit when storage is blocked), and the pre-paint theme script.

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### 👱 Muhammad Arslan (owner) · 2026-10-18 08:30 PKT

> okay, whhat is pending exactly . be brief

### 🤖 Claude · 2026-10-18 08:30 PKT

**Waiting on you now**

- Approve G4.3 (covers D-51).

**Build work left (Phase 4)**

- **Group 4, settings (S-6, S-7):** dark mode, the Theme select, both settings remembered across reloads.
- **Group 5, history (S-11, S-12):** saved calculations with Restore, Delete, Clear all and the confirmation dialog.
- **Group 6, layout and keyboard (S-5, S-8):** phone layout, tap targets, focus rings, tab order. It ends with a full `npm run verify` that must exit 0.

**After that**

- **Phase 5:** a full independent review; you tick the manual checklist on desktop and a phone, then decide each story's Status.
- **Phase 6:** the README, push, deploy to GitHub Pages, smoke tests on the live URL.

**Owner items still open**

- Create the public GitHub repo, by G4.6 at the latest so CI can run.
- Optional: `reuseExistingServer: false` in the frozen Playwright config (currently worked around with `CI=1`).
- Transcript exports into `transcripts/`.

**Tests still red:** 2 group 2 tests (AC-4.4, AC-9.2) wait on groups 4 and 5; most of the rest belong to groups 4–6.

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### 👱 Muhammad Arslan (owner) · 2026-10-18 08:30 PKT

> approved

### 🤖 Claude · 2026-10-18 08:30 PKT

G4.3 is approved. **Group 4: settings (S-6 theme, S-7 currency).** First I'll record the approval, then read the frozen specs for theme, currency, and the settings-related spec rules:

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 08:31 PKT

Next, the settings tests in `spec-rules.spec.ts` (fallback for corrupt settings, and the input-border contrast):

4 tool call(s): Bash, Read, Write ×2

### Claude · 2026-10-18 08:31 PKT

Next, the reducer gains `setTheme`, and takes its `Theme` type from the storage layer (UI may import from storage, not the other way round):

6 tool call(s): Read, Edit ×5

### Claude · 2026-10-18 08:32 PKT

Next, `AppContext` loads the settings before the first render, saves them only on a real user change, and applies the theme with a live `matchMedia` listener while "System" is selected:

4 tool call(s): Write ×3, Bash

### Claude · 2026-10-18 08:32 PKT

Next, the dark tokens, header layout, and the pre-paint script in `index.html`. That script applies the saved theme before React loads, so there's no flash of the wrong theme (spec §8.4).

4 tool call(s): Write ×2, Edit, Bash

### Claude · 2026-10-18 08:33 PKT

Clean. Now the group 4 tests in all three browsers. I'm adding S-4 (AC-4.4 should now pass) and the rest of group 2 as regression checks:

3 tool call(s): Bash ×3

### Claude · 2026-10-18 08:36 PKT

AC-6.1 passes in all three browsers. **Every group 4 test passes**, and group 2 shows no regressions. Next: build size, the D-52 record, and commits.

1 tool call(s): Bash

### Claude · 2026-10-18 08:37 PKT

Committed: `44c6afe`, `e6e639a` and `babc602`. Next, the independent review of group 4:

1 tool call(s): Agent

### Claude · 2026-10-18 08:37 PKT

The group 4 code is in and every group 4 test passes in all 3 browsers. The independent review is running now. Meanwhile, dark mode and the Theme select are live at http://localhost:5173/: try System, Light and Dark, and reload to check your choice is remembered.

### Claude · 2026-10-18 08:44 PKT

**0 blocking findings.** One Minor, G44-1, is a real bug of the same kind as the two-tab history problem (D-13): a second tab writes its stale theme back over yours. The fix is one line, applying D-13's re-read-before-write rule to settings, so I'll make it now.

4 tool call(s): Edit, Bash ×3

**Claude · 2026-10-18 08:46 PKT**

# Mini-gate G4.4: settings done

Group 4 (S-6 theme, S-7 currency) is built and independently reviewed. **Every group 4 test passes in all 3 browsers, and the reviewer found 0 blocking issues.** The commits are `44c6afe`..`0d163d4`.

**What was built**

- **Theme select** (System, Light, Dark), the dark-mode colours from the spec, and live following of the device setting while System is selected.
- **A pre-paint script** that applies the saved theme before the app loads. The reviewer confirmed there's no flash of the wrong theme.
- **Saved settings:** currency and theme are remembered across reloads. Corrupt saved values fall back field by field, and blocked storage just means they're kept for the visit, with no errors.

**Evidence**

| Check | Result |
|---|---|
| S-6, S-7 and the settings spec rules | All pass, 3 browsers (`CI=1`). I re-ran 66 of 66 after the fix. |
| Group 2 regressions | None. AC-4.4, red since group 2, now passes. |
| Still pending | AC-7.5 and AC-9.2, which need saved calculations (group 5) |
| Vitest, lint, format, typecheck | 294 of 294 tests passed; all other checks exit 0 |
| Bundle size | 83.3 KB of 150 KB gzipped |
| Freeze check | Only your two approved changes (D-46, D-48) |
| `code-reviewer` | **0 blocking**, 3 Minor |

The reviewer's own checks also came back clean:

- 10 kinds of corrupt settings, including a 2 MB string;
- storage blocked both ways;
- 30 rapid device-theme switches;
- dark-mode focus outline contrast of 7.43:1.

**Fixed after review:** G44-1. With two tabs open, changing the currency in one tab wiped out a theme chosen in the other. Settings are now merged over what's currently stored, and a two-tab check confirmed the fix. There's no automated test for this, because the tests are frozen.

**Corrections from me**

- My first test filter skipped AC-6.1 by accident (Playwright's `--grep` ignores case). I noticed, ran AC-6.1 separately (3 of 3 passed), and now always list what a filter excludes.
- I also briefly labelled an empty placeholder as a "two-tab probe". The real probe ran afterwards and is the one reported above.

**For you to approve:** D-52 (group 4 choices) and D-53 (how the Minors were handled).

**Try it** at http://localhost:5173/: switch the theme, reload, and switch your device between light and dark while System is selected.

Reply "approve G4.4" to start **group 5 (saved calculations: S-11, S-12)**. That covers the saved list, Restore, Delete, Clear all with its dialog, the two-tab rules and the storage-failure messages. It's the largest group left.

### 👴 Muhammad Arslan (owner) · 2026-10-18 13:37 PKT

> can you list down the features for the implementation now?

### 🤖 Claude · 2026-10-18 13:37 PKT

Here's where the features stand.

**Done (groups 1–4, all reviewed)**

- **Calculation:** monthly payment, total interest and total payable, using exact decimal arithmetic and the spec's re-amortising schedule. All reference loans match to the paisa.
- **Input handling:**
  - spaces are trimmed;
  - a trailing `%` on the rate is accepted;
  - western (`1,000,000`) and lakh (`10,00,000`) grouping are accepted, and any other comma use is rejected;
  - limits: 1,000 to 1,000,000,000; 0 to 50%; 1 to 40 years or 1 to 480 months.
- **Error messages:** shown under each field, exact text, announced to screen readers. Focus moves to the first invalid field, and old results are removed.
- **Stale results:** any edit hides old results and shows "Your inputs have changed…".
- **Reset:** clears the form, errors and results, keeps the settings, and moves focus to Loan amount.
- **Schedule:** month-by-month payment, interest, principal and balance. The header row stays visible while you scroll, the table scrolls by keyboard, and all 480 rows appear in about 150 ms.
- **Rounding note:** shown when individual payments differ by 0.01.
- **Currency:** PKR or USD, changing the symbol only, with a note saying so. Remembered across reloads.
- **Theme:** System, Light or Dark. It follows the device live, is remembered, and is applied before first paint. It also works when storage is blocked.
- **Error safety:** an error boundary shows a message instead of a blank page, and no console errors appear.

**Still to build**

- **Group 5, saved calculations (S-11, S-12):**
  - the newest 20 are auto-saved, newest first, skipping an exact repeat;
  - each entry shows three lines;
  - Restore refills the form, switches the currency and recalculates;
  - Delete with correct focus;
  - Clear all with a confirmation dialog;
  - two-tab safety;
  - messages when storage is blocked or corrupt, or a removal fails.
- **Group 6, layout and keyboard (S-5, S-8):**
  - final phone and desktop layout and 44 px tap targets;
  - focus rings everywhere;
  - the full tab order;
  - a keyboard-only end-to-end session.

Then a full `npm run verify`, which must pass.

- **Phase 5:** a whole-app review, and your hand check on desktop and a phone.
- **Phase 6:** the README, then deploy to GitHub Pages and smoke-test the live URL.

G4.4 is still waiting for your approval (D-52 and D-53) before I start group 5.

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### 👴 Muhammad Arslan (owner) · 2026-10-18 13:39 PKT

> can you plan to remove the limitation of loan amount interest rate and tenure as well

### 🤖 Claude · 2026-10-18 13:39 PKT

Before planning, I'll measure what removing the limits would actually do. The current limits (D-02) aren't arbitrary: the calculation's correctness guarantees were only verified inside them. Here's a quick run of the same algorithm outside them, in the scratchpad with Python `decimal`, nothing in the project:

🔧 4 tool call(s): Bash ×2, AskUserQuestion, Write

### 🤖 Claude · 2026-10-18 13:41 PKT

The bounds search is running. It checks large amounts against high rates and long tenures, small amounts against the same, and 600 random loans over the whole candidate range.

When it finishes, I'll bring you exact proposed bounds with the evidence for each, before any spec, AC or test changes.

G4.4 (D-52 and D-53) is still waiting for your explicit approval. You can give it together with your answer on the new bounds.

🔧 9 tool call(s): Read, Write ×2, Bash ×5, AskUserQuestion

### 🤖 Claude · 2026-10-18 13:46 PKT

While that runs, I'll record your D-55 decision:

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 13:46 PKT

D-55 is recorded (`aa6a34a`). The bounds check is still running under the new guarantee. It covers the smallest safe amount, the current limits, and two candidate ranges up to $10^{15}$, 100% and 1,200 months, with 20,000 loans each.

When it finishes, I'll bring you exact proposed limits with the evidence. Only after you approve them will I change the spec, ACs, copy, tests (through the test-designer) and code.

🔧 2 tool call(s): Read, Bash

### 🤖 Claude · 2026-10-18 13:49 PKT

This checks maximum rates of 50%, 60%, 75% and 90%, each with amounts from Rs 10 to $10^{15}$ and tenures up to 1,200 months (20,000 loans each). It takes about 5 minutes. I'll then propose the highest rate with zero failures, and bring you the full bounds proposal.

🔧 2 tool call(s): Read, Bash

### 🤖 Claude · 2026-10-18 13:54 PKT

The edge sweep is running: about 5,000 schedules for each candidate maximum. When it finishes, I'll give you the bounds proposal with its evidence for approval.

🔧 2 tool call(s): Read, AskUserQuestion

### Claude · 2026-10-18 13:56 PKT

Approved: **G4.4**, and the new limits with a **75%** maximum rate. First, I'll compute the new L-MIN and L-MAX reference figures with exact arithmetic, and confirm every guarantee holds for both:

2 tool call(s): Bash ×2

### Claude · 2026-10-18 13:57 PKT

The decisions are recorded. Before editing the spec, I'll pull its exact current wording for every line I need to change:

2 tool call(s): Bash ×2

### Claude · 2026-10-18 13:57 PKT

Now the spec edits. Each replacement asserts its exact old text first:

1 tool call(s): Bash

### Claude · 2026-10-18 13:57 PKT

The spec is updated; the only remaining "480" is the CSS `480px` table height, which is correct. Next, the ACs. Same approach:

2 tool call(s): Bash ×2

### Claude · 2026-10-18 13:57 PKT

The spec and ACs are updated (53 lines added, 35 removed, across the three docs). I'm committing the documents first, so the test-designer works from the approved text:

1 tool call(s): Bash

### Claude · 2026-10-18 13:58 PKT

The docs are committed (`8797b10`). Next, the test-designer updates the frozen tests to the new approved text. It's authorised by D-54 and D-55, and you'll see the diff.

1 tool call(s): Agent

### Claude · 2026-10-18 13:58 PKT

The test-designer is working through every test that depends on the old limits, the old L-MIN and L-MAX figures, the 0.01 guarantee and the old copy. It also updates the manual checklist.

I'm deliberately **not** changing the code yet. The updated tests should first fail against the old limits, which proves they test the new rule. Then I'll change `validate.ts` and `copy.ts` and make them pass.

### Claude · 2026-10-18 14:02 PKT

The test-designer is done. **30 domain tests now fail for the right reason**: the old limits reject the new valid values. Two of its notes need a decision from me:

- **#1 and #3** (a one-row loan counts as its own last row; "the 1,200th row" means 1,200 data rows) are sensible readings. I'll record them in D-56.
- **#4 is a real gap.** Nothing proves fast-check actually drew each exact bound. The deterministic grid of 24 bound combinations does cover all of them, though, so the bounds are tested either way.

Now I'll make the change in the code:

4 tool call(s): Bash ×3, Agent

### Claude · 2026-10-18 14:07 PKT

The `code-reviewer` is checking the limits change now. Its main jobs:

- check that every test change follows from the new approved text and nothing was weakened beyond D-55;
- fuzz the real code with its own random seeds over at least 20,000 loans.

The new limits are already live at http://localhost:5173/. Try the new L-MAX: `1000000000000000`, `75`, `100` Years.

When the review reports, I'll stop at this mini-gate before starting group 5.

### Claude · 2026-10-18 14:23 PKT

**0 blocking findings.** The reviewer's own fuzz of 20,010 loans against the real code found no counterexample: the final payment was off by more than 0.01 in 22 loans, never by more than 0.02, exactly as D-55 says.

It did catch a real miss of mine, **D54-1**: AC-2.15 still quotes the old message ("between 1,000 and 1,000,000,000"). My search didn't look for it. Before proposing the fix, I'll sweep every doc for anything else still quoting the old limits:

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 14:23 PKT

AC-2.15 is the **only** stale line left anywhere in the docs. I'll commit the review findings now, and change the AC text only once you approve it:

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 14:23 PKT

## Mini-gate: wider limits (D-54) and the corrected guarantee (D-55)

The change is built, tested and independently reviewed: **0 blocking findings**. The commits are `8797b10`, `16c8cc8`, `34a06b8`, `0c510f5` and `7c37819`.

**What changed**

- **New limits:** amount **10 to 1,000,000,000,000,000**, rate **0 to 75%**, tenure **1 to 100 years** (1,200 months).
- **Corrected guarantee:** every payment is within 0.01 of the headline, except the final one, which is within 0.02. The rounding note now reads "…differ by up to 0.02…".
- **New reference figures:** L-MIN and L-MAX recomputed and cross-checked with exact arithmetic.

**Evidence**

| Check | Result |
|---|---|
| Vitest | **297 of 297**, including all property tests at 1,000 runs over the new limits |
| E2E, groups 1–4 (group 5 and group 6 layout tests excluded, the exclusions listed) | **282 of 282**, all 3 browsers |
| Reviewer's own fuzz of the real code (seed 90210) | **20,010 loans, 0 counterexamples.** 22 final payments differed by 0.02, as D-55 allows. |

| Check | Result |
|---|---|
| Test integrity | Every test change is a hard-coded value from the new text. The only loosening is the final-row 0.02 that D-55 allows. Nothing was deleted or skipped. |
| L-MAX on screen | All 1,200 rows within 1 second in every browser; nothing clipped at 360 px |

**Your decisions**

1. **AC-2.15 still quotes the old message.** That's my miss: my search didn't include it. It's the only stale line left in any doc. May I change it to `Loan amount must be between 10 and 1,000,000,000,000,000.`? The tests already expect the new message.
2. **Approve D-56,** the test-designer's six readings. For example, a one-row loan counts as its own "last row", and "1,200th row" means 1,200 data rows.
3. **Noted for group 6:** at 200% zoom on a phone, the Calculate label clips and very long figures break mid-number (D54-2, D54-3).

Once you reply, I'll fix AC-2.15 and start **group 5 (saved calculations)**.

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### 👴 Muhammad Arslan (owner) · 2026-10-18 14:23 PKT

> yes please fix it

### 🤖 Claude · 2026-10-18 14:23 PKT

Fixing AC-2.15, recording the mini-gate, and committing:

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 14:24 PKT

AC-2.15 is fixed (`c9b960d`). **Group 5: saved calculations (S-11, S-12).** It's the largest group. First I'll read the frozen specs:

🔧 5 tool call(s): Bash ×2, Read ×2, Write

### 🤖 Claude · 2026-10-18 14:25 PKT

Next, the reducer gains history state and the restore transition:

🔧 2 tool call(s): Read, Write

### 🤖 Claude · 2026-10-18 14:26 PKT

Next, `AppContext` loads the history at start-up, alongside the settings:

🔧 3 tool call(s): Read, Edit ×2

### 🤖 Claude · 2026-10-18 14:26 PKT

Next, a successful Calculate now saves to history: re-read, skip a repeat, insert. If the save fails, history becomes unavailable for the visit:

🔧 9 tool call(s): Edit ×5, Write ×3, Bash

### 🤖 Claude · 2026-10-18 14:27 PKT

Lint and typecheck are clean. I'll run the group 5 tests in Chromium first for a fast loop, plus the cross-group tests that were waiting on history: AC-7.5, AC-9.2, and the AC-8.5 dialog focus rules.

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 14:28 PKT

**All 49 pass in Chromium on the first run.** Next, the same set in Firefox and WebKit, where focus and `storage` events behave differently:

🔧 5 tool call(s): Bash ×5

### Claude · 2026-10-18 14:39 PKT

Group 5 is committed: `ec76864`..`c0d02c3`. Next, the independent review:

🔧 1 tool call(s): Agent

### Claude · 2026-10-18 14:39 PKT

Group 5 is built and passes every test it covers in all three browsers. The whole suite is at **446 of 447**.

The `code-reviewer` is now checking group 5, including hostile storage cases, two real tabs, and keyboard-only deletes. I've also asked it to check independently my conclusion that the single Firefox failure is a test quirk.

Saved calculations are live at http://localhost:5173/: calculate a few loans, then try Restore, Delete and Clear all.

### Claude · 2026-10-18 14:54 PKT

**0 blocking findings.** The reviewer independently confirmed that the Firefox AC-8.1 failure is a test-portability quirk: its own neutral probe gave the same result, and the app's tab order is correct.

It left two failing `[REVIEW-n]` tests for real defects, both Minor. Before fixing, I'll read exactly what they expect:

🔧 12 tool call(s): Bash ×5, Read ×2, Edit ×3, SendMessage, PowerShell

### Claude · 2026-10-18 15:09 PKT

## Mini-gate G4.5: saved calculations done

Group 5 (S-11, S-12) is built and independently reviewed, with **0 blocking findings**. The commits are `ec76864`..`2aeaa66`.

**What was built**

- **Auto-save:** each successful calculation is saved; newest first, at most 20, an exact repeat skipped, each entry showing three lines.
- **Restore:** fills the form, switches and remembers the currency, recalculates, and leaves focus on the Restore button.
- **Delete:** removes by id, with focus moving to the next entry, else the previous one, else the heading.
- **Clear all:** asks for confirmation in a native dialog, with explicit focus at every step.
- **Two-tab safety:** every save and delete re-reads storage first, and the list follows changes made in other tabs.
- **Corrupt or tampered data:** dropped silently.
- **Failures:** blocked storage shows "unavailable"; a failed delete or Clear all keeps the entries and says so.

**Evidence**

| Check | Result |
|---|---|
| Full E2E suite (clean run, all 3 browsers) | **452 of 453**: the frozen suite plus 6 reviewer tests |
| Vitest, lint, format, typecheck | 297 of 297 tests passed; all other checks exit 0 |
| Freeze check | Only the authorised changes (D-46, D-48, D-54/55) |
| `code-reviewer` | **0 blocking.** Clean results from hostile stores (including 10,000 entries), interleaved two-tab writes, Escape pressed twice, keyboard-only deletes and a near-full storage quota. |

**Fixed after review** (D-58):

- **G45-1:** focus fell to the page body when another tab disabled Clear all mid-dialog. It now falls back to the heading.
- **G45-2:** a tampered entry could show the year "10000". The year is now checked in local time.
- **G45-3:** a tampered 10,000-entry store loaded slowly. Validation now stops after 20 valid entries.

**A slip of mine:** I ran Playwright while the reviewer was running it too, and the runs collided. The figures above are from a clean re-run, and the two runs won't overlap again.

**Your decision: the single Firefox failure (G45-4, AC-8.1)**

Both the reviewer and I proved with a page containing no app code that Playwright's Firefox **keeps focus on the last control when you Tab past it**, while Chromium and WebKit don't. The frozen test's line `expect(resetBtn).not.toBeFocused()` after that Tab is therefore testing a browser quirk. The app's tab order is correct, and the same test confirms the disabled Clear all is skipped.

🔧 1 tool call(s): AskUserQuestion

### Claude · 2026-10-18 16:23 PKT

Recorded: **G4.5 approved**, and **the portable assertion** for AC-8.1. The test-designer makes the change:

🔧 2 tool call(s): Bash, Agent

### Claude · 2026-10-18 16:24 PKT

The diff is exactly that one line, replaced by the portable check, and AC-8.1 now passes in all three browsers. Committing it with D-59 and the G4.5 approval:

🔧 7 tool call(s): Bash ×4, Read ×2, Write

### Claude · 2026-10-18 16:26 PKT

I'll also fix the comment in `ResultsSummary`: since D-48, tests find figures by label, not role. Then screenshot again at both widths:

🔧 7 tool call(s): Edit, Bash ×4, Read, Agent

### Claude · 2026-10-18 16:34 PKT

Group 6 is done, and `npm run verify` **exits 0 for the first time**: 453 of 453 E2E tests across the 3 browsers, and all 94 v1 ACs tested and passing.

The final Phase 4 review is running now. It does a full `verify` run of its own, checks the layout at six widths plus 200% zoom in both themes, a forced-colors (high-contrast) mode, and keyboard-only use in all three browsers. I'm not running any tests while it works, to avoid another collision.

One thing I need from you before Phase 4 can fully close: **the public GitHub repo**. AC-10.8 needs a passing run "on the CI test machine", and that can only happen once the repo exists and I push to it.

### 👴 Muhammad Arslan (owner) · 2026-10-18 16:37 PKT

> now, I want to try out new design, can you include detailed instruction on the frontend you have right now and what should the new design include, in order to fully implement it. Explain it to the design tool like someone will do to a designer and let the designer do all the creative work.

### 🤖 Claude · 2026-10-18 16:38 PKT

I'll write this as a design brief, addressed to the design tool the way you'd brief a designer. It covers what exists today, what's fixed and why, the states and screen sizes to design, and the deliverables, and it leaves the visual direction entirely to them. First, the setup for the document:

🔧 1 tool call(s): Artifact

### 🤖 Claude · 2026-10-18 16:38 PKT

I can't confirm that the Docs connector here is first-party, so I'll publish this as a standalone page instead:

🔧 1 tool call(s): Artifact

### 🤖 Claude · 2026-10-18 16:38 PKT

The page needs to be accurate about what the frozen tests check visually, since a designer can't break those. I'll pull the exact layout assertions from the tests:

🔧 3 tool call(s): Bash ×2, Write

### 🤖 Claude · 2026-10-18 16:41 PKT

Now I'll embed the two screenshots in the page and publish it:

🔧 2 tool call(s): Bash, Artifact

### 🤖 Claude · 2026-10-18 16:41 PKT

Written for: a designer or AI design tool that has never seen this project.

The brief is here: **EMI Calculator Design Brief**. It's private: it only opens for you until you share it from the page's Share menu, or you can copy the text straight into the design tool.

It reads like a brief to a designer: it explains the product and leaves all the creative decisions to them. It contains:

- **The ask and the people:** the three users, their situations, and what the design has to do for each.
- **What exists today:** two embedded screenshots, desktop light and phone dark.
- **The screen layout, every state to design** (15 rows), and **content extremes** using real strings, up to `Rs 75,000,000,000,000,003.34`.
- **What's fixed, and why:** the rules the tests enforce:
  - order and the side-by-side or stacked layouts;
  - exact copy;
  - 4.5:1 and 3:1 contrast;
  - a focus outline of at least 2 px;
  - 44 px tap targets;
  - system fonts only;
  - plain CSS;
  - the colour token names.

- **Where they have freedom,** and **what to deliver.** Ideally the deliverable is one plain-CSS stylesheet written against the existing class names.
- **Reference material:** the markup skeleton with classes, the current tokens, and every word on the screen.

Two things to know when you use it:

- **The new design must fit the existing constraints, or it comes back as proposals.** The brief tells the designer to list any text, markup or rule changes separately, so you can approve or decline each one, as we've done for everything else.
- **This is planning only.** When a design comes back, I'll implement it as its own change cycle: CSS against the frozen tests, a full `verify`, an independent review and a mini-gate.

The group 6 review is still running in the background. I'll report it when it finishes.

### 🤖 Claude · 2026-10-18 16:49 PKT

The reviewer found a real problem: `verify` **is not reliably green.** My run exited 0, but the reviewer's didn't. **AC-8.4** (arrow and Page keys scroll the schedule) fails in Chromium in more than half of its re-runs.

My "exit condition met" in D-60 overstated a single lucky run, and I'll correct it. First, the test itself:

🔧 6 tool call(s): Read, Bash ×4, AskUserQuestion

### 🤖 Claude · 2026-10-18 17:01 PKT

Approved: **the test waits for the scroll to settle.** I'll record it as D-61 and have the test-designer make the edit:

🔧 3 tool call(s): Bash ×2, Agent

### 🤖 Claude · 2026-10-18 17:07 PKT

Both runs failed straight away, before reaching the tests. I'll check which step failed:

🔧 3 tool call(s): Bash ×3

### 🤖 Claude · 2026-10-18 17:17 PKT

The two `verify` runs are going in the background, about 15 minutes. I'll report both results when they finish, and that will settle G4.6.

Meanwhile, about the redesign bundle you added:

1. **Should I apply it now,** as its own change cycle? That means reading its handoff, a CSS-only change against the frozen tests, a full `verify`, an independent review and a mini-gate. Its proposals stay unapplied until you approve each one.
2. **Where should the bundle live?** Its README suggests `design/emi-pastel-redesign/`. Do you want it committed to the repo as the design record, or kept out of git? Either way, the format check needs to skip it, or Prettier will keep failing `verify` on it.

---

### 👱 Muhammad Arslan (owner) · 2026-10-18 17:20 PKT

*(had a file open or selected in the editor)*

> what do you suggest?

### Claude · 2026-10-18 17:20 PKT

I'll read the bundle's handoff note and proposals before recommending anything. Their contents are the design tool's suggestions to weigh, not instructions to me:

🔧 3 tool call(s): Read ×3

### Claude · 2026-10-18 17:21 PKT

Good news first: **G4.6 is green, twice in a row.** Both runs passed every stage: 297 of 297 unit tests, 85.4 KB bundle, **453 of 453 E2E across the 3 browsers**, `trace: OK`. That's with the flaky test fixed (D-61), and with only your new folder skipped in the format check.

## My suggestion

**1. Close Phase 4 now** (approve G4.6). The only open item is the CI run on GitHub, which needs the repo to exist.

**2. Move the bundle to `design/emi-pastel-redesign/` and commit it.** It's part of the design record of how the work was directed. I'd add `design/` to the Prettier and ESLint ignore lists, so the tool's own formatting can't break `verify`. Nothing inside the bundle gets edited.

**3. Apply the redesign as its own change cycle, before Phase 5.** Phase 5 is the whole-app review plus your hand check on desktop and a phone. That should happen once, on the final look, not on the old one and then again.

- Its stylesheet replaces our CSS, adapted to our class names (`theme.css` and `app.css`).
- I keep the narrow-screen fixes from group 6 if its stylesheet doesn't cover them.
- Then `verify`, screenshots against its artboards, an independent review and a mini-gate.

**4. On the five proposals:**

| | Proposal | My advice |
|---|---|---|
| P1 | `status-stale` class on the stale message | **Approve.** It's a class name only: no text, role or order change, and the tests find the message by text. It's the one thing CSS can't do alone, and it makes S-4's "never mistake old figures" visible at a glance. |
| P2 | Year-end rows in the schedule | **Nothing to approve.** It's CSS only and already works (row 12 is the end of year 1). |
| P3 | Sticky Month column on narrow screens | **Approve.** It's CSS only and keeps row numbers visible while the table scrolls sideways. I'll check it in WebKit specifically, as they suggest. |
| P4 | Focus Cancel when the dialog opens | **Already done.** The app does this today (D-30), and AC-8.5 tests it. |
| P5 | Interest vs principal split bar | **Decline for v1.** It carries meaning, so it needs new copy, a new AC and new tests. It's a good backlog candidate. |

Reply with "go" (or adjust any line), and I'll record these as decisions and start with steps 1 and 2. I'd also suggest creating the GitHub repo soon. It's needed for CI, for the AC-10.8 check and for the live deploy in Phase 6.

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### Muhammad Arslan (owner) · 2026-10-18 17:22 PKT

*(had a file open or selected in the editor)*

> 1- Let's do it. 2-: Let's do it. 3- Let's do it. 4- Let's do it.

### Claude · 2026-10-18 17:22 PKT

All four approved. Step 1 is closing Phase 4, and step 2 is moving the bundle and committing it:

🔧 8 tool call(s): Bash ×5, Read ×2, Edit

### Claude · 2026-10-18 17:24 PKT

My replacement text also matched a second rule, `.history-card .field-error:empty`. The assertion stopped it before anything was written. I'll anchor the match to the start of the line:

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 17:24 PKT

The stylesheets are in place. Next, P1:

🔧 3 tool call(s): Edit, Bash ×2

### Claude · 2026-10-18 17:33 PKT

450 of 453. **3 WebKit failures**, all `locator.click` timeouts. That usually means something is covering the element being clicked. The details:

🔧 14 tool call(s): Bash ×9, Read ×4, Agent

### Claude · 2026-10-18 17:51 PKT

All four steps are done, and the redesign is now under independent review.

1. **Phase 4 is closed.** G4.6 is recorded, after two green `verify` runs in a row.
2. **The bundle is committed** in `design/emi-pastel-redesign/`, byte-identical to what you added. Prettier and ESLint skip it, so it can't fail `verify`.
3. **The redesign is applied** (`d808cc0`, `cd2f584`). `verify` exits 0: 297 of 297 unit tests and 453 of 453 E2E across the 3 browsers. Screenshots match the design's reference boards at 1280, 360 and 180 px, and nothing scrolls sideways.
4. **Proposals:** P1 and P3 are applied, P4 was already in place, and P5 is declined for v1 (D-63).

**Three deviations from the bundle** (D-64). Each protects a rule the designer couldn't know about:

- I removed `body { overflow-x: hidden }`, which would have hidden real overflow from the tests.
- I kept the empty error elements in the accessibility tree, so screen readers still announce new errors.
- I added the fatal-error style the handoff described but the bundle didn't include.

**Corrections and observations:**

- **A false alarm, corrected:** I briefly reported WebKit as 3× slower. A fair A/B test showed no real difference, so I retracted the claim (recorded in D-64).
- **The full E2E suite runs slower** with the new look, about 8.3 minutes against 5.5. It still passes. I haven't found the cause.
- **The design's mock-up shows two wrong figures for L-A** (`…185.46`). The app shows the correct `…185.45`.

**Try it now** at http://localhost:5173/: the new look is live in the dev server.

The reviewer is checking the contrast of text on the new pastel fills, the sticky Month column in WebKit, forced colours, and whether the focus ring that now appears after mouse clicks is acceptable. When it

reports, I'll fix whatever needs fixing and bring you the redesign mini-gate. After that comes **Phase 5**: the whole-app review and your hand check.

### Claude · 2026-10-18 18:02 PKT

**0 blocking findings.** The reviewer reproduced `verify` (exit 0, 453 of 453). It measured every text-on-pastel pair, with the lowest at 4.83:1, still above 4.5. It also confirmed my three deviations and the sticky Month column in WebKit.

It lists four Minors, RD-1 to RD-4, and I only have two from its message. Here are the other two from the findings file:

1 tool call(s): Bash

### Claude · 2026-10-18 18:03 PKT

## Redesign mini-gate

The pastel redesign is in and independently reviewed: **0 blocking findings**. Commits `2de0b0d`..`4063ea4`.

| Check | Result |
|---|---|
| `verify`, reproduced by the reviewer | **exit 0**: 297 of 297 unit tests, 85.4 KB, **453 of 453 E2E** across 3 browsers, `trace: OK` |
| Tests | Untouched by the redesign; only the authorised changes appear since the freeze |
| Contrast on the new pastel fills | All pass in both themes. The lowest is **4.83:1** (error text on the rose fill, light). |
| Focus, tap targets, overflow | 3 px outline everywhere. Nothing under 44×44 px. No sideways scroll at 180, 360, 768 or 1280 px, even with 1,200 rows. |
| Sticky header and Month column (P3) | Work in all 3 browsers, including WebKit |
| Against the design boards | Layout, colours, icons, the stale state and the dialog all match. The only differences are the boards' own wrong example numbers. |

**Minor findings**

- **RD-1, for your hand check:** a focus ring now appears after **mouse clicks** too, not just keyboard use. The designer did this deliberately, so focus is always visible.
- **RD-2:** the "!" icon on errors isn't read aloud. Confirmed in all 3 browsers.
- **RD-3:** in Windows high-contrast mode the monthly-payment tile and rounding note lose their fills, so only spacing separates them. Everything stays readable. There's an optional small CSS fix: borders shown only in high-contrast mode.
- **RD-4:** the bundle's contrast script wasn't run; the reviewer measured the live colours instead (above).

**Your decisions**

1. **Approve the redesign gate** (D-64).
2. **RD-3:** add the high-contrast borders? It's a few lines of CSS, and I'd recommend it. Or leave it as it is.

After that comes **Phase 5**. Per the loop plan, that's a whole-app review repeated until it reports 0 blocking issues. Then **you** tick the manual checklist in Chrome and in Safari or Firefox, on desktop and on a real phone, and decide each story's Status.

Phase 6 deploys to GitHub Pages, so the **public GitHub repo** is needed soon.

### 👴 Muhammad Arslan (owner) · 2026-10-18 18:03 PKT

> Can you add few visual (like charts diagram etc)?

### Claude · 2026-10-18 18:03 PKT

Yes, it's doable, but it's new scope, not styling. A chart carries meaning, so each one needs:

- a text equivalent for screen readers, which is new copy;
- its own ACs and tests through the test-designer;
- your approval at a gate.

Also, there's no charting library (no new dependencies), so the charts would be hand-built SVG in the app's own theme colours. That's actually ideal here: it's tiny, has no network calls, and works in both themes.

Pick the visuals you want, and I'll draft the story and ACs for your approval before building anything:

🔧 5 tool call(s): AskUserQuestion ×2, Skill, Bash, Write

### Claude · 2026-10-18 18:06 PKT

Approved, and I'll make exactly those four edits to the jobs file. Starting with the documents, beginning with that file, so you can see its diff on its own:

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 18:06 PKT

Exactly four lines changed, as approved. Next, the user stories: the S-19 section moves from the backlog into v1 with its new ACs, S-20 to S-22 are added, and the index is updated.

🔧 3 tool call(s): Bash ×3

### Claude · 2026-10-18 18:07 PKT

Confirmed: the spec and decision log are untouched. Only the jobs and stories edits from before are pending. Writing the script as a file:

🔧 6 tool call(s): Write ×2, Bash ×3, Edit

### Claude · 2026-10-18 18:10 PKT

The hooks are recorded (`77c223f`). Next, the test-designer writes the G3b tests for the 16 new ACs and the four domain functions:

🔧 1 tool call(s): Agent

### Claude · 2026-10-18 18:11 PKT

The visuals are underway, test-first. Done so far:

- **The proposal is approved and written into the docs** (`19d044b`). S-20, S-21 and S-22 are new stories, S-19 moves into v1, and the spec has the chart rules and copy. The jobs file has exactly the four edits you approved.
- **Throwing stubs** for the four new domain functions (`c1ce542`), so the new tests fail for the right reason.
- **D-66, the test hooks for the charts** (`77c223f`). This is proposed, for you to review at G3b. The charts are hidden from screen readers, which rely on the captions and tables, so tests need a few `data-` hooks to find hover areas and bar segments. I defined these before the tests were written, so nobody invents and freezes them by accident.

The **test-designer is now writing the G3b tests**: domain unit tests, a property test, and four new E2E specs. Only new files; every existing frozen test is untouched.

When it reports, I'll check that every new test fails for the right reason, then stop at **G3b** for your approval. That freezes the new tests under their own tag before I build the charts.

Still open from before: **the redesign gate** (D-64), and whether to add the high-contrast borders (RD-3).

### Claude · 2026-10-18 18:34 PKT

The test-designer reports every new test red for the right reason. I'll verify that myself, statically first: only new files, no frozen file touched, plus the counts.

🔧 3 tool call(s): Bash ×3

### Claude · 2026-10-18 18:39 PKT

Confirmed: all 4 are caused by `NOT_IMPLEMENTED`. For the E2E side, here's one Chromium run of the four new specs with the failure causes classified:

🔧 2 tool call(s): Bash ×2

### Claude · 2026-10-18 18:41 PKT

## Gate G3b: tests for the visuals written, all red for the right reason

**What was written** (by the test-designer; commits `892d518`, `52f75ad`, `1a70f9d`). All new files; no existing test, fixture or config changed.

- **48 Vitest tests** for the four new functions. They include 3 property tests at 1,000 runs, for example that the yearly totals always equal the loan's totals.
- **30 E2E tests per browser** across four new specs (S-19 to S-22), covering:
  - captions;
  - the yearly table values;
  - hover tooltips;
  - bar proportions;
  - contrast in both themes;
  - 360 px fit;
  - the 1-second budget with all charts present.
- **16 manual checklist lines.**

**Evidence** (my own checks)

| Check | Result |
|---|---|
| Vitest | **48 failed, 297 passed.** All 48 are the new tests, every one on a `NOT_IMPLEMENTED` stub; all 297 existing tests still pass. |
| E2E, Chromium | **30 of 30 fail.** 29 because no chart exists yet; 1 is AC-21.3's intended "charts didn't appear" timeout. **None passes early.** The test-designer reports 30 of 30 failing in Firefox and WebKit too. |
| Freeze, lint, typecheck | No frozen file touched; nothing skipped; no app code imported into E2E; lint and typecheck clean |
| `trace` | **OK**: 115 ACs, every v1 AC tested |

**Three gaps the test-designer found in my proposal. Please decide:**

1. **The yearly chart has no caption.** I forgot to define one, so the tests currently find that chart by its position. I suggest a fixed caption, `Interest and principal paid each year.`, which is new copy, and charts should be named. The tests don't need to change.
2. **The yearly tooltip includes the currency symbol,** like the other tooltips: `Year 1: principal Rs 338,857.21, interest Rs 422,945.63`. I suggest accepting this as written.
3. **Month counts in captions:** "over **1,200** months" or "over **1200** months"? I suggest `1,200`, matching the error message wording. The test accepts either.

The other 8 readings it reported are sensible, for example "a 0% loan may omit the interest segment entirely". I'll record them as D-67.

**To approve G3b:** say "approve G3b", plus your answers to 1 to 3 (or "as suggested"). I'll then:

- tag the freeze `g3b-visuals-tests-approved`;
- build the charts: domain functions, SVG components, and colours from the pastel tokens checked with the validator script;
- run `verify` and the independent review;
- bring you the mini-gate.

Still open: **D-64** (the redesign gate) and **RD-3** (high-contrast borders).

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### 👴 Muhammad Arslan (owner) · 2026-10-18 18:44 PKT

> approved

### 🤖 Claude · 2026-10-18 18:45 PKT

Approved. I'll take your suggestions for the three gaps as accepted, since you approved without separate answers, and record it that way. I'll update the spec and decision log, tag the freeze, and then build.

🔧 2 tool call(s): Write, Bash

### 🤖 Claude · 2026-10-18 18:45 PKT

Tagged: `g3b-visuals-tests-approved` is on `c62d9c2`. Now the build. The new frozen tests are the contract, so I'll read them first, starting with the shared helpers and the cost-bar spec:

🔧 2 tool call(s): Bash ×2

### 🤖 Claude · 2026-10-18 18:46 PKT

Next, the yearly and payment make-up specs:

🔧 4 tool call(s): Bash ×4

### 🤖 Claude · 2026-10-18 18:47 PKT

Light mode passes with lavender **#6A50D3** (the design's own focus colour) and coral **#C4512E**. Dark mode's candidates are slightly too light for the dark band, so I'll try a step darker:

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 18:47 PKT

I'm guessing at the band, so I'll read the exact dark-mode lightness band from the script rather than keep trying shades:

🔧 2 tool call(s): Bash ×2

### Claude · 2026-10-18 18:47 PKT

The domain expectations are clear. Implementing the four functions:

3 tool call(s): Write, Edit, Bash

### Claude · 2026-10-18 18:51 PKT

All **345 Vitest tests pass**, but `typecheck` rejects my `x.constructor.ROUND_HALF_UP` shortcut. I'll import the configured `Decimal` properly instead:

7 tool call(s): Bash ×2, Edit ×3, Write ×2

### Claude · 2026-10-18 18:56 PKT

Now the four chart components:

4 tool call(s): Write ×4

### Claude · 2026-10-18 18:57 PKT

Next, placing the charts in the results card, between the rounding note and the schedule (spec §8.3.2):

4 tool call(s): Read, Edit ×3

### Claude · 2026-10-18 18:57 PKT

The light tokens are in. For the dark ones, I'll read the reformatted `theme.css` first to place them exactly:

2 tool call(s): Bash ×2

### Claude · 2026-10-18 18:58 PKT

Next, the chart styles, appended to `app.css` using the design's tokens. The tooltip has `pointer-events: none`, so it never blocks the hover underneath it.

1 tool call(s): Bash

### Claude · 2026-10-18 18:58 PKT

Lint and typecheck are clean. Next, the four new specs in Chromium:

1 tool call(s): Bash

### Claude · 2026-10-18 18:59 PKT

**All 30 pass in Chromium on the first run.** Next, Firefox and WebKit:

1 tool call(s): Bash

### Claude · 2026-10-18 19:01 PKT

**90 of 90 across the 3 browsers.** The charts add rendering work to every calculation, which could affect the timing, layout and keyboard tests, so the real gate is the full `verify`:

1 tool call(s): Bash

### Claude · 2026-10-18 19:11 PKT

`verify` is running in the background, with the full suite now at 543 E2E tests. All 90 new chart tests already pass in Chromium, Firefox and WebKit, and all 345 unit tests pass.

Once `verify` reports, I'll commit the charts, record the implementation choices (D-68), and have the independent reviewer check them. That review covers the palette contrast in both themes, the 360 px and 200%-zoom layouts, hover in every browser, and the 1-second budget with all charts on screen.

The charts are already live at http://localhost:5173/ if you'd like a look. Calculate a loan and hover over the balance, payment and yearly charts.

🔧 3 tool call(s): Read, Bash ×2

### 🤖 Claude · 2026-10-18 19:14 PKT

While that runs, I'll commit the chart code. Its own 90 tests pass in every browser:

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 19:15 PKT

The charts are committed, one commit per story (`0d9967c`..`0dc743a`). I'm waiting on the second `verify` run and the isolated repeats of the WebKit test.

Once I know whether that failure is a one-off or a pattern, I'll report it either way. Then come the decision record (D-68), the independent review and the mini-gate.

---

### 👱 Muhammad Arslan (owner) · 2026-10-18 19:16 PKT

> [screenshot attached] [screenshot attached] can you also freeze the actual calculator on right , when you scroll the space on left is empty

### 🤖 Claude · 2026-10-18 19:16 PKT

You mean the **loan form** (the left card): keep it in view while you scroll through the results, instead of leaving that empty space. That's a CSS-only change and a good fit. My plan:

- **Desktop only (769 px and wider):** the form card becomes `position: sticky` and follows you down while the results scroll. Phones are unaffected, because there everything is one column.
- **Only when the window is tall enough to show the whole form.** Otherwise the bottom of the form, including Calculate, could end up out of reach on short laptop screens.
- **It stops at the end of the results,** so it never covers Saved calculations below.
- No markup, order or tab-order change, so the tests that depend on those aren't affected.

One thing to wait for first. My second `verify` run is still running in the background, checking whether that WebKit failure is a one-off. Changing the CSS mid-run would muddy that evidence, and so would running other browsers alongside it. As soon as it reports, I'll make the change, record it as D-69, and check it in all three browsers.

---

### 👱 Muhammad Arslan (owner) · 2026-10-18 19:24 PKT

> ok

### 🤖 Claude · 2026-10-18 19:24 PKT

I'll make the change as soon as the `verify` run reports. You'll get both results together: whether the WebKit failure repeated, and the sticky form checked in all three browsers.

🔧 7 tool call(s): Read, Bash ×4, Edit ×2

### 🤖 Claude · 2026-10-18 19:27 PKT

The code change applied, and lint and typecheck pass. The probe only failed to start, because I forgot to point Node at the project's modules this time. Re-running it:

🔧 4 tool call(s): Bash ×3, Write

### 🤖 Claude · 2026-10-18 19:28 PKT

D-68 and D-69 are recorded. They'll be committed together with the sticky-form code once `verify` passes.

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### 👱 Muhammad Arslan (owner) · 2026-10-18 19:35 PKT

> ok

### 🤖 Claude · 2026-10-18 19:36 PKT

I'll report as soon as `verify` finishes. Then I'll commit the sticky form with D-68 and D-69, and have the independent reviewer check the charts and the sticky form together before the mini-gate.

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### 👱 Muhammad Arslan (owner) · 2026-10-18 19:37 PKT

> ok

### 🤖 Claude · 2026-10-18 19:37 PKT

Still running: a full `verify` takes about 9 minutes, most of it the 543 E2E tests across three browsers. I'll post the result as soon as it lands.

🔧 3 tool call(s): Read, Bash, Agent

### 🤖 Claude · 2026-10-18 19:38 PKT

The sticky form is in and passes the full gate. The `code-reviewer` is now checking the charts and the sticky form, and it will reproduce `verify` itself. It covers:

- the money-never-a-number rule as the charts apply it;
- the palette contrast in both themes;
- L-MAX performance with all four charts;
- the sticky form at different window sizes, including whether keyboard focus ever ends up hidden;
- edge cases, from 1-month to 1,200-month loans, 200% zoom, and high-contrast mode.

You can try the sticky form now at http://localhost:5173/: calculate a loan in a full-size browser window and scroll. When the review reports, I'll fix what needs fixing and bring you the visuals mini-gate.

### 🤖 Claude · 2026-10-18 20:10 PKT

**0 blocking findings.** But the reviewer's `verify` did **not** pass: two WebKit two-tab tests timed out under full load. They're green alone and on the pre-charts commit, so the charts make WebKit just slow enough to trip them.

I'll fix all of it:

- **F-1 (Major):** at 180 px the yearly table pushed the page 129 px sideways. It gets a scroll wrapper that becomes keyboard-focusable **only when it actually overflows**. A permanent tab stop would break the frozen AC-8.1 tab order, and a scroll area you can't reach by keyboard is an accessibility failure.
- **F-2 (Major), and your open RD-3:** in high-contrast mode the cost bar and legend keys vanished in Chromium and Firefox. Data marks keep their colours there, with a system-colour outline. The same fix adds outlines to the headline tile and rounding note, which was RD-3. I'm including it as part of the fix, and you can still veto it.

- **F-4, the WebKit load:** the 1,200-row schedule and all four charts redrew on *every* state change, including the save to history right after Calculate. I'll memoise them so they only redraw when the loan or currency changes. That should take real load off WebKit.
- **F-3 tidy-ups:** the unused copy key, a duplicate import, hard-coded numbers, and the Year column becoming a row header.
- **F-5 (hover index not reset):** won't fix, because it can't be reached. I'll record why.

🔧 1 tool call(s): Bash

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*2026-10-18 20:12 PKT: the conversation reached the context limit and was compacted. Claude continued from a written summary, shown next.*

📋 **Summary carried into the next context**

This session is being continued from a previous conversation that ran out of context. The summary below covers the earlier portion of the conversation.

Summary:

1. Primary Request and Intent:
2. Muhammad Arslan, the product owner, is directing a client-only React + TypeScript **Loan EMI Calculator** for a job take-home. I write the code; he reviews and approves at gates.
3. CLAUDE.md rules, which must keep applying:
4. money is never a JS `number` (decimal.js only);
5. the algorithm in spec §5.2 is used exactly;
6. all visible text comes from `src/ui/copy.ts` and matches spec §7 exactly;
7. inputs are `type="text"`;
8. no new dependencies beyond spec §2 without asking (the pre-approved tooling list is in §2);
9. no network at runtime;
10. tests are the contract: after Gate G3, don't edit tests or the freeze scope of §12.4; if a test seems wrong, stop and explain to the owner;
11. never weaken, skip or delete tests;
12. **never change a story's Status without the owner's explicit approval**;
13. **never edit docs/app-roles.md, docs/jobs-to-be-done.md or ACs**: propose changes, which the owner edits or approves;
14. layer boundaries (domain imports nothing from ui or storage; storage imports nothing from ui);
15. tests titled `[AC-x.y]`;
16. small conventional commits;
17. a gate summary at each phase;
18. honesty: never claim more than the command output shows;
19. **record every decision in docs/spec/decision-log.md, append-only (D-24; owner: "keep all decision logs")**;
20. import `Decimal` only from `src/domain/money.ts`.
21. Workflow: phases with gates, defined in loop-plan.md.
22. Done: Phase 1 (G1), Phase 2 (G2), Phase 3 (G3, plus G3b for the visuals) and Phase 4 (groups 1 to 6; G4.6 closed). The redesign change cycle and the visuals change cycle are implemented.
23. Remaining: the visuals mini-gate, Phase 5 (whole-app review and the owner's manual checklist on desktop and phone, then the owner decides Statuses) and Phase 6 (README, push to GitHub, Pages deploy, smoke tests).
24. Most recent owner requests:
25. **visuals:** all four charts approved (D-65);

26. **sticky form:** "can you also freeze the actual calculator on right, when you scroll the space on left is empty" (D-69);
27. the owner replied "ok" to my status updates.

1. Key Technical Concepts:
   2. **Stack:** Node 24.16.0 (Windows, Git Bash plus PowerShell). Exact pinned versions:
      3. React 19.3.1, TypeScript 6.0.3, Vite 8.3.4, Vitest 5.0.5, jsdom 30.1.3;
      4. Playwright 1.64.1 (chromium, firefox, webkit), fast-check 4.10.2, decimal.js 10.6.0 (configured clone: precision 40, ROUND_HALF_UP);
      5. ESLint 9.39.5 with typescript-eslint, react-hooks and jsx-a11y; Prettier 3.9.10.
   6. `npm run verify` = lint → format:check → typecheck → test → build → size (< 150 KB gzipped) → test:e2e → `trace --results`.
      7. **Always run Playwright with** `CI=1`, because a stale preview server on 4173 can otherwise be reused.
      8. Never run two Playwright suites at once.
      9. Probe scripts go in the scratchpad, run with `NODE_PATH=<repo>/node_modules`.
   10. **Algorithm:** re-amortising schedule; interest multiply-first (D-37). Guarantee (D-55): every payment is within 0.01 of the headline except the final one, which is within 0.02.
   11. **Limits (D-54):** amount 10 to 10^15; rate 0 to 75%; tenure 1 to 100 years or 1 to 1,200 months.
   12. **DOM contract** (spec §8.3.1, D-43/D-48/D-66):
      13. summary figures are found with `getByLabel`;
      14. charts are a `<figure>` named by its `<figcaption>`, with the SVG `aria-hidden`;
      15. D-66 hooks: `[data-hit-area]` (item k of n at (k − 0.5)/n of the width), `[data-segment]`, `[data-series]`;
      16. tooltips have `role="tooltip"`.
   17. **Freeze tags:** `g3-tests-approved` (authorised later changes: 749144e D-46, 5751988 D-48, 16c8cc8 D-54/55, 04de524 D-59, 95259bc D-61) and `g3b-visuals-tests-approved` (c62d9c2).
   18. **Design:** the "Soft ledger" pastel redesign from the bundle in `design/emi-pastel-redesign/` (kept as received; excluded from Prettier and ESLint).
   19. **Chart palette** (validated with the dataviz validator script): light principal #6A50D3, interest #C4512E; dark principal #957EEC, interest #D4703C. Balance and loan use the principal colour.

1. Files and Code Sections:
   2. **docs/spec/decision-log.md:** D-00 to D-69, append-only. The latest:
      3. D-65: visuals approved;
      4. D-66: test hooks, approved at G3b;
      5. D-67: G3b approved, the three copy gaps closed with my suggestions, and the test-designer's readings;
      6. D-68: visuals implementation, **Proposed** for the visuals mini-gate;
      7. D-69: sticky form, approved request, implementation for review.
      8. Still Proposed or open: D-64 (redesign deviations) and RD-3 (forced-colors borders).
   9. **docs/user-stories.md:**
      10. S-19 moved from Backlog to v1, with AC-19.1 to 19.4;
      11. new S-20 (AC-20.1 to 20.6), S-21 (AC-21.1 to 21.3) and S-22 (AC-22.1 to 22.3);
      12. 22 stories and 115 ACs; every Status is still "Not implemented".
   13. **docs/jobs-to-be-done.md:** exactly four owner-approved edits: J-2 + S-20, J-5 + S-22, J-8 + S-21, and J-9's S-19 no longer "(backlog)".
   14. **docs/spec/technical-spec.md:**
      15. §1 scope;
      16. §3.1.1 summary.ts API and formatPercent;
      17. §7 chart and yearly copy (including `yearly.chart.caption` "Interest and principal paid each year.", grouped `{n}`, tooltips with currency);
      18. §8.3.1 contract rows;
      19. §8.3.2 chart rules;

20. §16.
21. **docs/process/proposal-visuals.md:** the approved visuals proposal.
22. **src/domain/summary.ts** (new): ``ts export interface YearRow { year: number; months: number; interest: Decimal; principal: Decimal; balance: Decimal } const HUNDRED = new Decimal(100) const round1 = (x: Decimal) => x.toDecimalPlaces(1, Decimal.ROUND_HALF_UP) export function costSplit(loan: CalculatedLoan) { const interestPercent = round1(loan.totalInterest.times(HUNDRED).div(loan.totalPayable)); return { interestPercent, loanPercent: HUNDRED.minus(interestPercent) } } export function interestShare(payment: Decimal, interest: Decimal): Decimal { if (payment.isZero()) return new Decimal(0); return round1(interest.times(HUNDRED).div(payment)) } export function yearlySummary(loan): YearRow[] // groups of 12 rows; exact sums; balance = group's last balance ``
23. **src/domain/format.ts:** ``ts import { Decimal } from './money' export function formatPercent(x: Decimal): string { const rounded = x.toDecimalPlaces(1, Decimal.ROUND_HALF_UP); return ${rounded.isZero() ? '0.0' : rounded.toFixed(1)}% } ``
24. **src/ui/copy.ts:**
    25. `chart.cost` (title, summary(pct), loan, interest);
    26. `chart.balance` (title, summary(amount, zero, months, count) with month/months, tooltip(month, balance));
    27. `chart.makeup` (title, summary(first, last), tooltip(month, principal, interest));
    28. `chart.series` (principal, interest);
    29. `yearly` (title, chartCaption, cols(currency), tooltip(year, principal, interest)).
30. **src/ui/components/charts/chartGeometry.ts:**
    31. `VIEW_W = 1000` and `VIEW_H = 100`;
    32. `ratio(value, max)`: Decimal division, then `toDecimalPlaces(6).toNumber()`; used only for geometry;
    33. `stepBandPath(top, bottom?)`: one stepped-area path;
    34. `barsPath(top, bottom, fill = 0.64)`;
    35. `centrePercent(k, n)`;
    36. `useBandHover(count)`: pointermove sets `index = floor(f × count)`;
    37. `groupCount(n)`.
38. **src/ui/components/charts/ChartParts.tsx:**
    39. `Legend({ first })`: HTML `ul.chart-legend` with swatches;
    40. `AxisEnds({ count })`: numbers only, "1" and n;
    41. `Tooltip`: `div role="tooltip"` with `left: x%` and `translateX(-x%)`;
    42. `HoverBand`: a rect with hard-coded 1000/100 (reviewer F-3).
43. **The four chart components** (in `src/ui/components/charts/`):
    44. **CostBreakdown.tsx:** h3 title; `figure.chart` containing `div.cost-bar` with `span[data-segment=loan][data-series=loan]` and, only when the interest is non-zero, `span[data-segment=interest][data-series=interest]` (flexGrow = the percentages); legend; figcaption.
    45. **BalanceChart.tsx:** has a duplicate import of `../../../domain/money` (F-3). An SVG `.chart-plot` with `preserveAspectRatio="none"`, `aria-hidden`, `data-hit-area` and the hover handlers, plus `path.mark-principal[data-series=balance]`.
    46. **PaymentMakeupChart.tsx:** a principal path plus an interest path stacked on it.
    47. **YearlySummary.tsx:** h3 `#yearly-title`; figure with the bars SVG, legend and caption; then `table.yearly-table[aria-labelledby=yearly-title]` with `td` cells (F-3 suggests a `th` row header).
48. **src/ui/components/ResultsPanel.tsx:** after RoundingNote, renders `CostBreakdown`, `BalanceChart`, `PaymentMakeupChart`, `YearlySummary`, then `ScheduleTable`. The stale message uses `className="status status-stale"` (P1).
49. **src/ui/App.tsx:** ``tsx <main className="layout"> {/ *No semantics: bounds the sticky form to the form/results row (D-69).* /} <div className="layout-main"><LoanForm /><ResultsPanel /></div> <HistoryPanel /> </main> ``
50. **src/ui/theme.css:** the redesign tokens, plus `--chart-principal` and `--chart-interest` in the light, dark and fallback-media blocks.

51. **src/ui/app.css:**
    52. the redesign CSS with the D-64 deviations (removed `body overflow-x: hidden`; `.field-error:empty::before { content: none }`; a `.fatal` style);
    53. `.layout` is a single column, and `.layout-main` is the 4fr/6fr grid (both single column at ≤ 768 px);
    54. the sticky rule: ``` `css @media (min-width: 769px) and (min-height: 820px) { .form-card { position: sticky; top: var(--space-6); } }` ```
    55. section 14, charts: `.chart-block`, `figure.chart`, `.chart-plot { height: 140px }`, `.mark-*`, `.chart-hover-band`, `.chart-axis`, `.chart-legend`, `.swatch`, `.chart-tooltip { pointer-events: none }`, `.cost-bar { display: flex; gap: 2px }`, `.cost-segment`, `.yearly-table` (not a scroll region), and a ≤ 560 px media block.
56. **New test files** (frozen at g3b):
    57. src/domain/summary.test.ts and src/domain/format-percent.test.ts;
    58. e2e/visual-helpers.ts;
    59. e2e/s19-yearly.spec.ts, s20-cost.spec.ts, s21-balance-chart.spec.ts and s22-makeup.spec.ts.
60. **Reviewer files:**
    61. e2e/review/g45.spec.ts (REVIEW-1/2, passing);
    62. **e2e/review/visuals.spec.ts** (new; [REVIEW-V1] and [REVIEW-V2], currently failing);
    63. docs/process/review-findings.md (findings sections through "Visuals").
64. **Recent commits:** a5ee7e7 (domain); 0d9967c, fd97fa6, e30a41d, 0f656e4, 323f1f3, 0dc743a (charts); a19ea7f (sticky); 4333476 (D-68/D-69 docs). The branch is `v1`. There is no GitHub remote yet.

1. Errors and fixes:
    2. **A stale vite preview on port 4173 served an old build:** I stopped it and use `CI=1` for every E2E run (D-49).
    3. **Playwright's `--grep` is case-insensitive** and an exclusion filter silently dropped AC-6.1: I now list excluded tests with `--list`.
    4. **I labelled an empty probe placeholder as a real "two-tab probe":** I corrected it and ran the real probe.
    5. **My Playwright runs collided with the reviewer's** (ENOENT trace files): no concurrent runs from then on.
    6. **False "WebKit 3× slower" claim:** an alternating A/B disproved it, and I retracted it in D-64.
    7. **Bash heredoc parse failure on a large Python script:** I wrote scripts to scratchpad files instead (d65_docs.py, d67_docs.py, d68_docs.py).
    8. **A chain broke because `grep -c` returns exit 1 on a count of 0:** I re-ran the remaining steps.
    9. **A `/tmp` script couldn't find `@playwright/test`:** moved to the scratchpad and set NODE_PATH.
    10. **TypeScript error on `x.constructor.ROUND_HALF_UP`:** imported `Decimal` from `./money` instead.
    11. **The `.field-error:empty` replacement matched twice:** anchored it with a newline.
    12. **`mv` of the bundle was denied** (a file was open in the IDE): copied it byte-identically, then removed the original.
    13. **Prettier `format:check` failed on the design bundle:** added `design/` to .prettierignore and to ESLint `globalIgnores` (D-62).
    14. **The sticky form overlapped Saved calculations:** added the `.layout-main` wrapper, so the sticky containing block ends at the results.
    15. **Flaky AC-8.4** (Chromium smooth-scroll race): an owner-approved test fix, D-61.
    16. **Firefox-only AC-8.1 tab-past-the-end quirk:** an owner-approved portable assertion, D-59.
    17. **Recurring WebKit 8 s or 30 s click timeouts under full-suite load:** they pass in isolation, and verify passed on re-runs. The latest reviewer run had 2 §9.3 two-tab WebKit failures (F-4), which passed 40/40 on later runs and with one worker.

1. Problem Solving:
    2. **Visuals:** implemented and committed. Verify passed with 543/543 E2E on my runs (the second and the post-sticky runs).
    3. **Reviewer's latest findings** (BLOCKING 0):
        4. **F-1 Major:** at 180 px (200% zoom) after L-B or L-MAX, the page scrolls sideways by 129 px, because `table.yearly-table` can't shrink. [REVIEW-V1] is red in all 3 browsers.

5. **F-2 Major** (the open RD-3): in forced-colors mode in Chromium and Firefox, the cost-bar segments and legend swatches become Canvas with no border, so they're invisible. [REVIEW-V2] is red in Chromium and Firefox.
6. **F-3 Minor:**
7. unused `copy.chart.cost.interest` (the legend uses `copy.chart.series.interest`);
8. duplicate money import in BalanceChart;
9. HoverBand hard-codes 1000/100;
10. the Year column is a `td`, not a row header.
11. **F-4 Minor:** WebKit §9.3 two-tab load flake, seen only at HEAD.
12. **F-5 Minor:** the hover index isn't reset when the loan changes; not reproducible.
13. The reviewer also suggested making "ratio and percentages may be JS numbers for geometry" an explicit carve-out in the spec.

1. All user messages:
2. "Can you go through the whole folder and share your thoughts?"
3. "1- should be fixed, 2- we will do at end"
4. "what questions do you have for me right now?"
5. The list answering items 1 to 13: CLAUDE.md formatting yes; this is the planning phase; keep 2 dp; keep AC-12.6 and add a dialog line; reword AC-8.5; items 6 to 12 approved; "Let's create a repo then." Also: "Right you are going to come up with solid plan, You will implement everything in phases story by story."
6. AskUserQuestion answers: local git now, GitHub later; identity Muhammad Arslan <arslan8954@gmail.com>.
7. "one other thing, keep all decision logs"
8. "ok"
9. "okay start phase 1"
10. "seems good, let's move to phase 2"
11. "okay run the app"
12. ";et" (unclear; I asked for clarification)
13. "let's do phase 3"
14. "approved" (G3)
15. "next step?"
16. "Can you please restart the app?"
17. "approved" (G4.1)
18. AskUserQuestion answers: AC-9.2 fix "Expect $ lines"; continue building.
19. AskUserQuestion: D-43 fix "Yes, use getByLabel".
20. "okay let's move on" (G4.2)
21. "okay, whhat is pending exactly . be brief"
22. "approved" (G4.3)
23. "can you list down the features for the implementation now?"
24. "can you plan to remove the limitation of loan amount interest rate and tenure as well"
25. AskUserQuestion answers: wide verified limits, now, before group 5. D-55: option A. New limits with max rate 75%, plus approve G4.4.
26. "yes please fix it" (AC-2.15)
27. AskUserQuestion: G45-4 portable assertion; approve G4.5.
28. AskUserQuestion: AC-8.4 "Wait for scroll to settle".
29. "now, I want to try out new design, can you include detailed instruction on the frontend you have right now and what should the new design include, in order to fully implement it. Explain it to the design tool like someone will do to a designer and let the designer do all the creative work."
30. "what do you suggest?"
31. "1- Let's do it. 2-: Let's do it. 3- Let's do it. 4- Let's do it."
32. "Can you add few visual (like charts diagram etc)?"

33. AskUserQuestion: all 4 visuals. Proposal "Approve as written"; jobs file "You make exactly those edits".
34. "approved" (G3b)
35. Screenshots plus "can you also freeze the actual calculator on right , when you scroll the space on left is empty"
36. "ok"
37. "ok"
38. "ok"

1. Pending Tasks:
   2. Fix the reviewer's visuals findings, in CSS or markup where approved:
      3. **F-1:** the yearly table must not cause horizontal page scroll at 180 px. For example, allow it to shrink or wrap (smaller padding and font, `overflow-wrap`, `table-layout`), or put it in a wrapper with `overflow-x: auto`. Careful: no new tab stops, because AC-8.1 is frozen. A scroll wrapper would need to be focusable, which would change the tab order, so prefer making the table shrink.
      4. **F-2/RD-3:** add `@media (forced-colors: active)` rules so the cost segments and swatches stay visible, for example borders or `forced-color-adjust: none` on the marks, or system colours with patterns.
      5. **F-3 cleanup:** remove the duplicate import; use VIEW_W and VIEW_H in HoverBand; optionally a `th scope="row"` for Year (the test reads `th, td`, so that's allowed); handle the unused `copy.chart.cost.interest`.
      6. **F-5:** optionally reset hover on loan change.
   7. Then re-run `[REVIEW-V1]` / `[REVIEW-V2]` and the full `CI=1 npm run verify`; record a decision (D-70) for how the findings were handled, plus the money-as-ratio carve-out in the spec if the owner agrees; commit the reviewer's findings file and visuals.spec.ts; present the **visuals mini-gate**: approve D-68 and D-69, and the outstanding **D-64** (redesign gate) and **RD-3**.
   8. Later:
      9. **Phase 5:** whole-app review until 0 blocking; the owner's manual checklist; the owner decides Statuses.
      10. **Phase 6:** README, push, deploy, smoke tests, `trace --results`, transcripts.
      11. The owner must create the **public GitHub repo** (needed for CI, the AC-10.8 CI timing, and the Pages deploy).
      12. Optionally, decide on `reuseExistingServer: false` in the frozen playwright.config.ts.

1. Current Work:
   2. I had just received the code-reviewer's visuals mini-gate report: BLOCKING FINDINGS 0.
      3. Its `CI=1 npm run verify` exited 1, with 2 WebKit §9.3 two-tab timeouts (F-4). Those passed 40/40 on re-runs, and with one worker.
      4. The two Majors are F-1 (180 px overflow from `table.yearly-table`, test [REVIEW-V1]) and F-2 (forced-colors invisible cost bar and swatches, test [REVIEW-V2]).
      5. The Minors are F-3 and F-5.
      6. New file: e2e/review/visuals.spec.ts. docs/process/review-findings.md has a "Visuals" section appended.
   7. Neither the reviewer's files nor any fixes are committed yet.
   8. Dev server: running at http://localhost:5173/ (background task b8saqnfbu).

1. Optional Next Step:
   2. Address the visuals review findings, then present the visuals mini-gate. I had told the owner: "When the review reports, I'll fix what needs fixing and bring you the visuals mini-gate."
   3. Concretely:
      4. commit the reviewer's findings and the `e2e/review/visuals.spec.ts` tests;
      5. fix F-1, the yearly table overflowing at 180 px, without adding tab stops;
      6. fix F-2 with forced-colors rules for the cost segments and legend swatches;
      7. clean up F-3;
      8. run `CI=1 npx playwright test e2e/review/visuals.spec.ts`, then a full `CI=1 npm run verify` with nothing else running;
      9. record D-70 in the decision log;

10. commit;
11. present the mini-gate summary: verify results, the honest F-4 flake note, and the decisions D-68, D-69, D-64 and RD-3 for the owner to approve.

If you need specific details from before compaction (like exact code snippets, error messages, or content you generated), read the full transcript at: C:\Users\arslan.NINEALPHA\.claude\projects\c--Users-arslan-NINEALPHA-muhammad-arslan-calculator\9d763f75-d2e9-4d5a-b730-c05dd7d4b1f5.jsonl Continue the conversation from where it left off without asking the user any further questions. Resume directly — do not acknowledge the summary, do not recap what was happening, do not preface with "I'll continue" or similar. Pick up the last task as if the break never happened.

The messages after this summary are the most recent messages from before compaction, kept verbatim. The summary was written without seeing them, so something it says has not happened yet may already have happened in them.

### Claude · 2026-10-18 20:12 PKT

Starting the fixes. First I'll read the review tests and the components I'm changing.

3 tool call(s): Bash ×3

### Claude · 2026-10-18 20:12 PKT

Next I'll measure what is actually wide at 180 px before choosing the fix.

1 tool call(s): Bash

### Claude · 2026-10-18 20:13 PKT

The ESM loader ignores NODE_PATH, so I'll switch the probe to CommonJS.

7 tool call(s): Bash ×7

### Claude · 2026-10-18 20:14 PKT

I'll make the code changes now: the scroll wrapper, the row header, the legend labels, the hover guard, and memoisation.

2 tool call(s): Bash ×2

### Claude · 2026-10-18 20:14 PKT

Next: the yearly scroll wrapper (F-1) and the row header, then memoising the results details.

1 tool call(s): Bash

### Claude · 2026-10-18 20:15 PKT

The table's header cell styles would now also hit the new Year row headers, so I'll scope them to `thead`. Then I'll add the wrapper and forced-colors CSS. First, RD-3's exact wording:

5 tool call(s): Bash ×5

### Claude · 2026-10-18 20:17 PKT

The React hooks lint rule objects to reading the ref off the returned object during render. I'll destructure it instead.

1 tool call(s): Bash

### Claude · 2026-10-18 20:17 PKT

Lint and typecheck are clean. Next, the targeted E2E run in 3 browsers, covering the reviewer's tests, the four visual specs and the keyboard spec.

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 20:19 PKT

129 of 129 pass, including [REVIEW-V1] and [REVIEW-V2] in all 3 browsers. Next I'll confirm that the yearly table is a tab stop only when it overflows, by probing at 180 px and at 1280 px.

🔧 3 tool call(s): Bash ×3

### Claude · 2026-10-18 20:20 PKT

Next, three commits: the reviewer's work, the fixes, and the decision record.

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 20:20 PKT

Now the full `CI=1 npm run verify`, with nothing else running. It takes about 10 minutes.

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 20:21 PKT

I've fixed the reviewer's findings and committed them, and the full `CI=1 npm run verify` is running now (about 10 minutes). The visuals mini-gate summary will follow once it finishes.

**What changed:**

- **F-1 (Major), page scrolling sideways at 180 px:** the yearly table now sits in its own scroll area. It's a named, keyboard-focusable region only while the table is too wide to fit. Shrinking the table couldn't work: its four headers need 284 px, and the column is 130 px wide.
  - Checked in all 3 browsers: at 180 px the page doesn't scroll sideways and the region appears. At 1280 px it's a plain block, so the frozen AC-8.1 tab order is unchanged.
- **F-2 (Major), plus your open RD-3, in high-contrast mode:** the cost bar and legend keys keep their colours and get an outline. The headline tile and rounding note also get borders; that part is RD-3, and you can veto it.
- **F-4, the WebKit timeouts under load:** the charts and the 1,200-row schedule now redraw only when the loan or currency changes, and moving the pointer over a chart no longer recomputes it. This may help, but it doesn't prove the timeouts are gone; the verify run will show that.
- **F-3 tidy-ups:** the unused copy key, the duplicate import, the hard-coded numbers, and the Year column is now a row header.
- **F-5:** I added a small guard so the chart ignores a stale hover position. The reviewer couldn't trigger the bug.

**Checks so far:**

- lint, typecheck and format are clean, and the unit tests pass (345 of 345);
- the reviewer's two tests plus the S-19 to S-22 and keyboard tests pass, 129 of 129 across the 3 browsers.

No test or frozen file was touched. Everything is recorded as **D-70 (Proposed)** in decision-log.md.

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 20:30 PKT

The full verify passed on its first run. I'll add that result to D-70 as a dated line, since the log is append-only, and commit it.

🔧 1 tool call(s): Bash

**Claude · 2026-10-18 20:30 PKT**

# Visuals mini-gate (S-19 to S-22, plus the sticky form)

The full `CI=1 npm run verify` **exited 0 on its first run**, with every test passing in Chromium, Firefox and WebKit. The reviewer found 0 blocking issues; the two Majors it raised are now fixed.

## 1. What was done

- **Charts (S-19 to S-22, D-68):** the cost breakdown bar, balance over time, payment make-up, and the yearly chart and table. All 16 of their ACs pass in all 3 browsers.
- **Sticky form (D-69, your request):** on wide, tall windows the form stays in view while you scroll the results, and stops before Saved calculations.
- **Review fixes (D-70):**
  - **Page scrolled sideways at 180 px (F-1):** the yearly table scrolls inside its own area instead. It becomes a keyboard tab stop only while it's too wide to fit, so the normal tab order (AC-8.1) is unchanged.
  - **Charts vanished in high-contrast mode (F-2, and your open RD-3):** the cost bar and legend keys keep their colours and get an outline. The headline tile and rounding note also get borders; that part is RD-3.
  - **WebKit timeouts under load (F-4):** the charts and schedule no longer redraw on unrelated changes, such as the history save right after Calculate.
  - **Tidy-ups (F-3, F-5):** an unused copy key now used, a duplicate import, hard-coded numbers, the Year column as a row header, and a guard on a stale hover position.

## 2. Evidence

| Check | Result |
|---|---|
| `CI=1 npm run verify` | **exit 0** |
| lint / format / typecheck | clean |
| Vitest | 345 / 345 |
| Bundle size | 87.4 KB of 150 KB |
| E2E (3 browsers) | **549 / 549** in 7.9 min (543 before, plus the reviewer's 2 new tests in each browser) |
| `trace --results` | OK; every v1 AC tested and passing |
| 180 px probe, 3 browsers | page overflow 0; the table is a region there, and not at 1280 px |

No frozen test or config was touched. Commits: `6c5edee` (reviewer files), `61e51b9` (fixes), `bf060cd` and `c5d197a` (D-70).

## 3. Unsure about

- **F-4:** one clean run doesn't prove the WebKit timeouts are gone. Earlier runs failed only intermittently.
- **My fixes haven't been independently reviewed.** The reviewer's own two tests now pass, but its agent hasn't read the new code. I suggest the Phase 5 whole-app review covers it, rather than another review round now.

- Every story Status is still "Not implemented", as it should be until you've checked them by hand in Phase 5.

## 4. Not done

- **GitHub CI:** still blocked, because the remote doesn't exist yet. AC-10.8's "on the CI test machine" can't be proven until it does.

## 5. Questions for you

1. Do you **approve D-68** (how the charts were built) and **D-69** (the sticky form)?
2. Do you **approve D-70**, including the RD-3 borders on the headline tile and rounding note, or should I drop those borders?
3. Do you **approve D-64**, the redesign's three deviations from the design bundle? It has been waiting since the redesign gate.
4. **Spec wording:** the reviewer suggested saying explicitly in §8.3.2 that drawing ratios and percentages may become JS numbers, for chart geometry only. Money values never do. Should I add that line?
5. If you approve, I'll start **Phase 5**: the whole-app review, then your hand check on desktop and phone.

---

### 👱 Muhammad Arslan (owner) · 2026-10-18 21:53 PKT

> rerun the server

🔧 4 tool call(s): ToolSearch, TaskStop, Bash ×2

### 🤖 Claude · 2026-10-18 21:54 PKT

The dev server is running again at **http://localhost:5173/** and answering requests. I stopped the old one first, and the new one runs the latest code, including the D-70 fixes.

Things you can check in the browser:

- **Sticky form:** on a wide, tall window, scroll the results and the form should stay in view.
- **Yearly table at 200% zoom:** set the window narrow or zoom to 200%, and the yearly table should scroll inside its own area without the page moving sideways.
- **High-contrast mode:** turn on Windows Contrast themes, and the cost bar and legend keys should stay visible with outlines.

The five gate questions from my last message are still open whenever you're ready.

🔧 0 tool call(s):

### 🤖 Claude · 2026-10-18 21:54 PKT

The new dev server exited with an error, so the `up` I reported may have come from the old process still dying. Checking:

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 21:54 PKT

Stopping the old task ended its shell, but its Vite child process (PID 23668) survived and is still holding port 5173. I'll confirm what that process is before stopping it:

🔧 1 tool call(s): PowerShell

### Claude · 2026-10-18 21:54 PKT

It's the old Vite dev server from this project. I'll stop it and start a fresh one.

🔧 3 tool call(s): PowerShell, Bash ×2

### Claude · 2026-10-18 21:55 PKT

The dev server has restarted at **http://localhost:5173/**. Vite printed its "ready" line, and it's a new process listening on the port.

My last message said it was already running, and that was wrong. The `up` I saw came from the old server: stopping the old task had ended its shell, but its Vite process kept running and held the port. The new server then failed with "Port 5173 is already in use". I checked that the old process was this project's Vite server, stopped it, and started a new one.

The five gate questions are still open whenever you're ready.

---

### 👱 Muhammad Arslan (owner) · 2026-10-18 21:55 PKT

> what are the gates?

### Claude · 2026-10-18 21:56 PKT

A **gate** is a stop point at the end of each phase. I stop, show you the evidence, and wait for your "approved" or your corrections before going on. That keeps you deciding, while I build and the subagents test and review.

## The six gates

| Gate | Ends phase | What you approve | Status |
|---|---|---|---|
| **G1** | 1. Spec challenge | The requirements, after the spec critic attacked them | ✅ Approved |
| **G2** | 2. Scaffold | The empty project, tooling, CI files and traceability script | ✅ Approved |
| **G3** | 3. Tests first | The tests, written before the code; they are then frozen | ✅ Approved (G3b for the chart tests) |
| **G4.1–G4.6** | 4. Build | One mini-gate per story group (domain, core UI, schedule, settings, history, layout) | ✅ All approved |
| **G5** | 5. Independent review | Whole-app review until 0 blocking findings, **plus your hand check** on desktop and phone; you decide each story's Status | ⏳ Next |
| **G6** | 6. Release | README, push to GitHub, live site, smoke tests | ⏳ Later |

Your later requests (the redesign and the charts) each ended in an extra **mini-gate** of the same kind.

## What's waiting on you now: the visuals mini-gate

1. **D-68**: approve how the four charts were built?
2. **D-69**: approve the sticky form?
3. **D-70**: approve the review fixes, including the extra high-contrast borders on the headline tile and rounding note (RD-3)? Or should I drop those borders?

4. **D-64**: approve the redesign's three deviations from the design bundle? This one has been waiting since the redesign gate.
5. Add one line to the spec saying chart geometry may use plain JS numbers for ratios, while money never does?

A reply like "approve all" works, or answer them one by one. After that comes **Phase 5**: the whole-app review, then your hand check.

One thing to plan for: you'll need to create the public GitHub repo before Phase 6. CI and the live site depend on it.

---

### 👱 Muhammad Arslan (owner) · 2026-10-18 21:56 PKT

> 1-Approved 2-Not working 3-Approved 4-Approved 5-Approved

### 🤖 Claude · 2026-10-18 21:57 PKT

Thanks. Approved D-68, D-70 (including the RD-3 borders), D-64, and the spec line. I'll record those once the sticky form is fixed.

The sticky form doesn't work for you. My guess is the height rule: it only turns on when the window is at least **820 px tall**. A typical laptop browser window is about 650 to 750 px tall, and even the spec's own desktop test size (1280 × 800) falls under that. I'll measure before changing anything.

🔧 5 tool call(s): Bash ×4, Edit

### 🤖 Claude · 2026-10-18 21:58 PKT

That one was an unsettled scroll: once it settles, WebKit holds the form at 24 px at every scroll position. Next, the full `verify`, because this CSS affects every desktop-width test.

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 21:58 PKT

While verify runs, I'll record your answers. That's documentation only; no source files change while the tests run.

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 21:59 PKT

The sticky form is fixed, and the full `verify` is running now (about 8 minutes). I'll commit once it passes.

**What was wrong:** my rule only turned sticky on when the browser window was at least 820 px tall. The form itself is only 544 px. Most laptop windows are 650 to 800 px tall, so for you it never stuck. My own check had used a tall window, so I missed it. That was my mistake.

**The fix:** whenever form and results sit side by side, the form sticks. If a window is too short to show the whole form, the form card scrolls on its own, so Calculate stays reachable.

**Checked in Chromium, Firefox and WebKit:**

- **1280 × 800 and 1366 × 650:** the form stays 24 px from the top while you scroll, and stops just above Saved calculations.
- **1280 × 480:** the form card scrolls on its own and stays pinned.
- **768 px and narrower:** stacked as before, not sticky.

**Recorded** (not yet committed):

- your approvals of D-64, D-68 and D-70 (including the RD-3 borders);
- the new spec line on chart drawing numbers, as D-71 in spec §8.3.2;
- under D-69, that it failed your check, why, and the fix.

The dev server at http://localhost:5173/ already has the fix. Reload the page and scroll the results.

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### 👱 Muhammad Arslan (owner) · 2026-10-18 22:02 PKT

> https://github.com/iarslandogar/Loan-calculator
>
> new repo

### Claude · 2026-10-18 22:02 PKT

Thanks, that unblocks CI and the live site. The plan already has me add the remote and push once you created the repo (D-22). First I'll check the repo is empty and whether anything in our workflows hard-codes the old name, `muhammad-arslan-calculator`.

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 22:02 PKT

The repo is **public and empty**, and the workflows read the repo name automatically, so the Pages path will be `/Loan-calculator/` with nothing to change. Next I'll check what a push would publish.

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 22:02 PKT

The push is safe to make:

- **125 tracked files:** source, tests, docs and the design bundle. No `node_modules`, `dist`, reports or logs.
- **Secret scan:** nothing found.

**What will happen when I push:**

- `main` has only the two planning commits and no workflows, so pushing it **won't deploy anything**. The live site comes in Phase 6, when `v1` is merged.
- `v1` (79 commits) will start the **CI workflow on GitHub's Linux machines**. That gives the CI evidence AC-10.8 needs, which we haven't had until now.
- The two freeze tags (`g3-tests-approved` and `g3b-visuals-tests-approved`) go up too, so the reviewer's freeze check works from the repo.

The local `verify` for the sticky fix is still running. I'll wait for it, commit the fix and your gate answers, and then push.

One small note: `package.json` still uses the name `muhammad-arslan-calculator`. It's cosmetic and doesn't affect the build or the Pages path, so I'll leave it unless you want it renamed.

---

### 👱 Muhammad Arslan (owner) · 2026-10-18 22:03 PKT

> cool, now we need to deploy it

### Claude · 2026-10-18 22:03 PKT

Deploying means merging `v1` into `main`: the deploy workflow runs on every push to `main`. That is earlier than the plan's order, which put Phase 5's review and your hand check before release. It's your call; I'll record it as a decision, and Phase 5 can still run against the live site.

First, what the deploy workflow will do:

1 tool call(s): Bash

### Claude · 2026-10-18 22:03 PKT

The pipeline is sound: pushing to `main` runs the full `verify` on GitHub's Linux machines first. If anything fails, nothing deploys. If it passes, the workflow publishes to Pages and then runs the `@smoke` tests against the live URL.

**One step only you can do** (GitHub doesn't let the workflow do it). Please do it now, before I push `main`:

> **Repo → Settings → Pages → Build and deployment → Source: "GitHub Actions"**
> https://github.com/iarslandogar/Loan-calculator/settings/pages

Without it, the deploy job fails at "configure-pages".

**My steps, in order:**

1. Wait for the local `verify` of the sticky fix, which is still running, then commit it with your gate answers.
2. Push `v1` and the two freeze tags. CI runs on `v1` and proves the build on Linux.
3. Merge `v1` into `main`, record it as D-72 (deploying before Phase 5, at your direction), and push `main`. That starts verify → deploy → smoke.
4. Report the live URL, **https://iarslandogar.github.io/Loan-calculator/**, and the workflow results.

Tell me when the Pages setting is done. I'll do steps 1 and 2 as soon as verify finishes.

---

### Muhammad Arslan (owner) · 2026-10-18 22:06 PKT

> [screenshot attached] I see this

### Claude · 2026-10-18 22:06 PKT

That's right: **Source shows "GitHub Actions"**, so the setting is done. Ignore the suggested workflows (Next.js, Hugo and so on); the repo already has its own deploy workflow.

Now the local verify result:

1 tool call(s): Bash

### Claude · 2026-10-18 22:06 PKT

Verify is still running: 405 of 549 E2E tests done. Checking for failures so far:

1 tool call(s): Bash

### Claude · 2026-10-18 22:06 PKT

The Pages setting is done.

Local verify has one failure so far, at 405 of 549 E2E tests: **[webkit] [AC-1.10]**, after 11.1 s. That looks like the known WebKit timeout under full load (the 8 s action limit), not the sticky change: AC-1.10 only types a rate

with a "%" and reads the result. I'm not assuming that, though. When the run finishes I'll read the actual error and re-run the test alone in WebKit.

Push plan, honestly stated:

- **This run won't exit 0.** I'll push `v1` only if the failure is the load timeout and every other test passed.
- **GitHub CI then decides.** Its `verify` on Linux is what gates the deploy, so a real failure there blocks publishing.

I'll report back as soon as the run finishes.

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 22:09 PKT

Verify **failed**: 547 passed and 2 failed, both in WebKit. The second is more serious than a timeout: **AC-10.8 is the 1-second performance limit.** Reading the details:

🔧 7 tool call(s): Bash ×5, Read, Edit

### 🤖 Claude · 2026-10-18 22:12 PKT

While it runs I'll record the fix as **D-72**, and the early deploy as **D-73**.

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 22:12 PKT

Not pushed yet: the last full `verify` failed, and I found the cause. A re-run with the fix is in progress, about 9 minutes.

**What failed:** 547 tests passed and 2 failed, both in WebKit.

- **[AC-1.10]:** a click timed out under load, the same intermittent failure we've seen before.
- **[AC-10.8]:** this one was real. Drawing all 1,200 rows of the largest loan took **1,133 ms against the 1-second limit**.

**The cause was my D-70 fix, not the sticky form.** The new yearly-table check forced the browser to lay out the whole page right after Calculate, inside the measured time. In WebKit alone, 8 runs each:

| Variant | Median |
|---|---|
| Current code | 223 ms |
| Sticky change removed | 274 ms (sticky isn't the cause) |
| **Forced check removed** | **114 ms** |

Under the full test suite's load, that doubled cost went over 1 second. I removed the forced check; the browser's own resize notification does the same job after layout. The yearly table still scrolls at 180 px and isn't a tab stop at normal widths, checked in all 3 browsers.

The earlier "549 of 549" green run had hidden this cost, so from now on timing-sensitive changes get a measured comparison. It's recorded as **D-72**, and deploying before Phase 5 as **D-73**.

**If this verify passes**, I'll commit, push `v1` with the tags, then merge into `main` and push. That push runs verify again on GitHub's Linux machines, then the deploy, then the smoke tests on the live site. If it fails, I'll report that rather than push.

🔧 2 tool call(s): Bash ×2

### Claude · 2026-10-18 22:27 PKT

AC-10.8 now passes in all 3 browsers. I also found a leftover `vite preview` on port 4174 from my A/B probe; it's idle, but I'll stop it. Next, the four errors:

🔧 2 tool call(s): Bash ×2

### Claude · 2026-10-18 22:30 PKT

Only **2 of 12 passed**. That isn't load. Something is really failing, so I'll look at the full summary:

🔧 3 tool call(s): Bash ×3

### Claude · 2026-10-18 22:35 PKT

Still failing with the old rule, so **the sticky change isn't the cause**. Before blaming the code, I'll check the machine for stuck browser processes left over from the earlier runs and probes:

🔧 3 tool call(s): PowerShell, Bash ×2

### Claude · 2026-10-18 22:37 PKT

Animation frames run normally (about 54 fps in WebKit), so that's not it. Instead of guessing further, I'll run one failing test alone and read the page snapshot Playwright saves on failure:

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 22:38 PKT

Alone it passes, in 29 s including the build. So it fails only when 4 WebKit workers run at once, and earlier today 4 workers were fine. Let me see what is using the CPU:

🔧 1 tool call(s): PowerShell

### Claude · 2026-10-18 22:39 PKT

Only about 20% CPU is in use now, mostly your Chrome and VS Code, so the machine isn't overloaded. I'll run the four tests in parallel once more and read a failure snapshot this time:

🔧 4 tool call(s): Bash ×3, PowerShell

### Claude · 2026-10-18 22:41 PKT

Everything is committed locally (3 new commits). Nothing has been pushed yet: I said I wouldn't push on a failing verify without telling you first, and the local verify isn't green.

**Where things stand:**

- **AC-10.8 is fixed.** With the forced layout removed, it passed in all 3 browsers even on the slow machine.
- **The remaining failures look environmental, not the code:**
  - the laptop is on battery at 37%, and the CPU is throttled;
  - the exact code that passed 549 of 549 earlier today now fails the same four WebKit tests;
  - all four pass when run alone, and the page snapshots show the app correct, just slow;
  - all are timeouts, in WebKit only.

**Two ways forward. I recommend doing both:**

1. **Push `v1` now.** It doesn't deploy anything; it only starts CI on GitHub's Linux machines. The spec names that run as the 3-browser evidence that counts (NFR-1), and it's the run AC-10.8's "on the CI test machine" needs. I'd merge to `main`, which is what deploys, **only after that CI run is green**.
2. **Plug the laptop in**, and I'll re-run the full local `verify` to confirm the timeouts were the battery.

Shall I push `v1` now?

### 👴 Muhammad Arslan (owner) · 2026-10-18 22:41 PKT

> go

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 22:42 PKT

`v1` is pushed. Next, the two freeze tags, and `main` as it is now (just the two planning commits, which contain no workflows, so nothing deploys):

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 22:42 PKT

Everything is on GitHub. The repo's default branch is currently `v1`, because it was pushed first; that's harmless, and it can be switched to `main` later. Now the CI run:

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 22:42 PKT

CI is running on `v1` (run 39150790709). I'll watch it in the background; a full verify on Linux takes roughly 10 to 15 minutes.

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 22:42 PKT

Pushed, and the first CI run is under way. Nothing is deployed yet.

**On GitHub** (https://github.com/iarslandogar/Loan-calculator):

- `v1` holds all the work. Pushing it started **CI run #1**, which runs the full `verify` (all 3 browsers) on GitHub's Linux machines: https://github.com/iarslandogar/Loan-calculator/actions/runs/39150790709
- `main` has only the two planning commits and no workflows, so nothing deploys from it yet.
- **The two freeze tags** are up, so anyone can check that no frozen test changed after approval.

**Next:** I'm watching the run in the background. If it's **green**, I'll merge `v1` into `main` and push. That runs verify again, then deploys to **https://iarslandogar.github.io/Loan-calculator/**, then runs the smoke tests against the live site. If it's **red**, I'll read the logs and tell you what failed before doing anything else.

Small notes:

- **The repo's default branch is** `v1` for now, because it was pushed first. After the merge you can switch it to `main` under Settings → General, or I'll remind you then.
- **If you can, plug the laptop in.** It doesn't affect the GitHub run, but I still want one clean local `verify` to confirm the battery theory.

### 👴 Muhammad Arslan (owner) · 2026-10-18 22:47 PKT

> pluged in

### 🤖 Claude · 2026-10-18 22:47 PKT

Thanks. I'll confirm the power state, then start the full local `verify`. It won't clash with the GitHub run, which is on GitHub's machines.

🔧 2 tool call(s): PowerShell, Bash

### Claude · 2026-10-18 22:47 PKT

Two runs are going now:

1. **The local `verify` on mains power** (about 8 to 10 minutes). This tests the battery explanation. If it's green, the four WebKit timeouts were the throttled CPU. If not, I'll dig into the code again.
2. **GitHub CI on `v1`** (Linux, all 3 browsers). This one decides whether we deploy.

I'll report each as it finishes. The merge to `main`, which is the deploy, waits for a green GitHub run.

🔧 2 tool call(s): Bash ×2

### Claude · 2026-10-18 22:52 PKT

The push didn't finish within 2 minutes, which is unusual given the `v1` push took seconds. Checking:

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 22:53 PKT

GitHub's `main` hasn't moved, so the push is stuck. Checking whether a sign-in window is waiting for you:

🔧 1 tool call(s): PowerShell

### Claude · 2026-10-18 22:53 PKT

**Git Credential Manager has opened a "Select an account" window and is waiting for you.** The push is paused until you pick an account.

Please find that window (it may be behind others or flashing in the taskbar) and choose your **iarslandogar** GitHub account. The push to `main` then continues by itself, and the deploy starts:

1. verify on GitHub's Linux machines;
2. deploy to https://iarslandogar.github.io/Loan-calculator/;
3. the smoke tests against the live site.

I'll watch the deploy run once the push goes through. The local verify is still running in parallel.

🔧 1 tool call(s): Bash

### Claude · 2026-10-18 22:58 PKT

**Local verify passed on mains power:** exit 0, 549 of 549 E2E tests across 3 browsers, 345 of 345 Vitest, `trace` OK. That confirms it: the earlier WebKit timeouts came from the CPU throttling on battery, not from the code.

The push to `main` is still waiting on the **"Select an account"** window from Git Credential Manager; GitHub's `main` hasn't moved yet. Once you pick your **iarslandogar** account there, the deploy starts, and I'll follow it through to the live smoke tests.

🔧 2 tool call(s): Bash ×2

### Claude · 2026-10-18 23:23 PKT

The push went through after you picked the account: GitHub's `main` is now at `182e87d`, the same commit as `v1`.

The **Deploy run** has started: https://github.com/iarslandogar/Loan-calculator/actions/runs/39153885811. It has three steps:

1. **verify** on Linux, the same check that just passed on `v1`, about 12 minutes;
2. **deploy** to GitHub Pages;

3. **smoke**: it waits until the live site serves this exact commit, then runs the `@smoke` tests against **https://iarslandogar.github.io/Loan-calculator/**.

I'm watching it in the background and will report when it finishes. Don't trust the live URL until the smoke step is green; until then the site may be missing or stale.

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 23:32 PKT

The run succeeded overall. My filter cut off the smoke job's own result line, so I'll check that job explicitly rather than infer it:

🔧 2 tool call(s): Bash ×2

### 🤖 Claude · 2026-10-18 23:33 PKT

**The app is live at https://iarslandogar.github.io/Loan-calculator/.** The deploy run passed all three steps: the full verify on Linux, the publish to GitHub Pages, and the smoke tests against the live URL. I also opened the live page: it loads, and it serves the exact commit we tested.

**Also confirmed:**

- **The battery explanation:** plugged in, the local verify passed every test, 549 of 549. The earlier WebKit timeouts were the throttled CPU.
- **AC-10.8 on a CI machine:** the 1-second test passed on GitHub's Linux machines, the first CI evidence we have. I couldn't read the per-test numbers, because GitHub only shows job logs to a signed-in account.
- **The decision log** has this as D-73, pushed to `v1` only, so it didn't trigger another deploy.

**Still to do:**

- **Default branch:** it's `v1` for now. Please switch it to `main` under Settings → General → Default branch, so visitors land on the released code.
- **Phase 5:** the reviewer agent goes over the whole app, and you check it by hand, which you can now do on your phone with the live URL. After that you decide each story's Status; they're all still "Not implemented".
- **Phase 6:** the README. The repo has none yet, and the assignment requires one.

Shall I start Phase 5 with the whole-app review?

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### 👨 Muhammad Arslan (owner) · 2026-10-18 23:56 PKT

> can you create a transcript for the whole development we did?

### 🤖 Claude · 2026-10-18 23:56 PKT

Yes. Claude Code keeps a raw log of every session on this machine, and I'll build the transcript from those. First, what logs exist:

🔧 1 tool call(s): Bash

### 🤖 Claude · 2026-10-18 23:56 PKT

Everything is in **one main session log (43.7 MB)** plus **19 subagent logs** (the spec critic, test designer and reviewer runs). I'll check the log format before writing a converter:

🔧 12 tool call(s): Bash ×8, Write, Edit ×3

### Claude · 2026-10-18 23:59 PKT

All 40 of your messages are present now, including the one sent mid-task. The "system-reminder" count went *up*, to 9, so I'll check what those are:

🔧 4 tool call(s): Bash ×3, Write

### Claude · 2026-10-19 00:00 PKT

A quick look at the opening of the main transcript, to check that it reads well:

🔧 1 tool call(s): Bash

### Claude · 2026-10-17 20:44 PKT

```
</details>

### 🤖 Claude · 2026-10-19 00:00 PKT

Written for: the people reviewing your take-home, who need to see how you directed the work.

The transcript of the whole development is ready in [transcripts/](transcripts/). Nothing is committed or
pushed yet, so you can look first.

**Read these:**
- [transcripts/README.md](transcripts/README.md) is the index. It explains the folder, gives a timeline
linking each phase and gate to your own words, points out moments worth reading (the frozen tests
challenged properly, the false rounding guarantee caught, claims corrected), and says what was left out.
- [session-01-claude-code.md](transcripts/session-01-claude-code.md) is the full readable session (1.1
MB).
  - It runs from 17 Oct 20:44 to 18 Oct 23:59, Pakistan time.
  - All **40 of your messages are verbatim**, including the one you sent while I was mid-task, plus my
237 replies.
  - The 646 tool calls are collapsed and shortened, and your 10 answers to my multiple-choice questions
are in full.
  - The one context compaction is marked, with the summary I continued from.
- [subagents/](transcripts/subagents/) has the 19 spec-critic, test-designer and reviewer runs, each
linked from the point where it started.
- [raw/](transcripts/raw/) has the filtered session logs (17 MB). [make_transcripts.py]
(transcripts/make_transcripts.py) is the script that built everything from Claude Code's local logs.

**What I left out:**
- **Claude Code's internal request records (about 30 MB).** They hold the tool's system prompt and a list
of your other connected accounts (Apollo, Atlassian, ClickUp, Google Drive and others). If you want the
fully unfiltered logs published anyway, say so and I'll add them.
- **My private reasoning blocks**, and the long middles of tool outputs. Each cut says how much was
removed, and the filtered raw logs keep the full tool outputs.
- **Secrets:** I scanned all logs for tokens and keys and found none.

**Before this is complete:**
- **`session-00-requirements`**, the claude.ai chat where the requirements were drafted, has to be
exported by you (D-00e); I can't reach it.
- **Optional:** you can also run `/export transcripts/session-01-export.md` in this chat to add Claude
Code's own export alongside mine.

Shall I commit and push this to `v1`? Pushing `v1` doesn't trigger a deploy.

## 👴 Muhammad Arslan (owner) · 2026-10-19 00:03 PKT

> yes

### 🤖 Claude · 2026-10-19 00:03 PKT

I'll regenerate first, so the transcript includes this last exchange too, and record the filtering choice
as **D-74**:
```

```
<<TOOLS>>🔧 1 tool call(s): Bash
```